Qualitative content analysis: theoretical foundation, basic procedures and software solution

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1. Introduction: Research Methods Between Qualitative and Quantitative Paradigms

This introduction criticizes the methodological dichotomization of qualitative and quantitative research, defines Qualitative Content Analysis as a mixed methods approach (containing qualitative and quantitative steps of analysis) and advocates common research criteria for qualitative and quantitative research. Finally, a step-by-step model of the (qualitative-quantitative) research process is presented.

Perhaps, no issue in social sciences contains more differences of opinion than research methodology. And there is perhaps no topic with more importance for scientific work and valid research results than that of adequate research methods. The disagreement about methods between different social science disciplines becomes evident in different forms: In sociology, an interpretive field study orientated tradition and a quantitative survey oriented tradition coexist. In psychology, quantitative experiments for causal inferences are within mainstream whereas qualitative approaches only occur recently. In economics, case studies were predominant at the time when quantitative economics rose. “This plurality makes it difficult to establish criteria for evaluation or to design curricula for teaching research methods” (Packer, 2011, p. 2). More and more, method preferences seem to be individual and arbitrary decisions of researchers.

1.1 Science War: Conflicting Paradigms

In 1959, Snow diagnosed two cultures in sciences, working with different methods: a constructivist, postmodern position and a realistic position (Snow, 1959). In the nineties, after a parody on postmodern constructivism (the “Sokal hoax”) the situation exacerbates to a science war (Ross, 1996; Bucchi, 2004). On the one hand stands a rigid positivistic conception of research with a quantitative, experimental methodology, on the other hand an open, explorative, descriptive, interpretive conception using qualitative methods.

Two factors have recently intensified the methodological debate in social sciences: under the flag of “evidence basement” the requirement for experiments in the form of Randomized Controlled Trials (RCTs) has been formulated as the only valid scientific procedure. Not only within health studies (evidence based medicine) but as well in education, social work and other social sciences, RCTs are seen as gold standard and institutions have been founded to collect, to review and to meta-analyze such studies (Cochrane Collaboration, Campbell Collaboration, cf. www.campbellcollaboration.org). This development has mobilized qualitative researchers. Denzin (2010) published a qualitative manifesto (“A call to arms”), connecting the evidence-based movement with neoliberal politics, using a narrow model of objectivity, opposed against another form of science as tentative, interpretive (the researcher as bricoleur), as well as critical, empowerment-guided (the researcher as actor), following not only scientific criteria but also poetic and artistic criteria (embodied experience, narrative truth, research report as literary text).

If not coming from a position of radical constructivism (treating different positions as equivalent subjective constructions), this situation is extremely unsatisfying for experienced researchers and newcomers. Of course the question of adequate research methods needs a deeper discussion of positions in theory of science (e.g. realism versus constructivism) of course. This could hardly be done within the framework of this book.

Excurse: A Theory of a Science Framework for Qualitative Content Analysis

Guba and Lincoln (2005) are differentiating between four paradigms in the theory of science. The following table characterizes the basic beliefs of those approaches:

Table 1: Basic beliefs (metaphysics) of alternative inquiry paradigms (Guba & Lincoln, 2005, p. 193)

Item	*Positivism*	*Postpositivism*	*Critical Theory*	*Constructivism*
Ontology	Naïve realism – "real" reality but apprehendible	Critical realism – "real" reality but only imperfectly and probabilistically apprehendible	Historical realism – virtual reality shaped by social, political, cultural, economic, ethnic, and gender values; crystallized over time	Relativism – local and specific constructed and co-constructed realities
Epistemology	Dualistic/objectivistic; findings true	Modified dualistic/objectivistic; critical tradition/community; findings probably true	Transactional/subjectivist; value-mediated findings	Transactional/subjectivist; created findings
Methodology	Experimental/manipulative; verification of hypotheses; chiefly quantitative methods	Modified experimental/ manipulative; critical multiplism; falsification of hypotheses; may include qualitative methods	Dialogical/dialectical	Hermeneutical/dialectical

I will try to discuss those positions on the background of our context of content analysis. If we are looking at approaches to text analysis, we can differentiate between two extreme positions, coming from different epistemological backgrounds:

- The hermeneutical position, embedded within a constructivist theory, tries to understand the meaning of the text as interaction between the preconceptions of the reader and the intentions of the text producer. Within the hermeneutical circle (cf. chapter 3.2) the preconceptions are refined and further developed in confrontation with the text. The result of the analysis remains relative to the reading situation and the reader.

- The positivistic position tries to measure, to record and to quantify overt aspects of the text. Those aspects of the text can be detected automatically; their frequencies can be analyzed statistically. The results of the analysis claim objectivity.

A strict contraposition of those positions ignores the possible convergences: The social constructivist theory formulates the possibility of an agreement between different individual meaning constructions and allows by that the concept of a socially shared quasi-objective reality. Modern hermeneutical approaches try to formulate rules of interpretation. By this, the analysis gains objectivity. On the other hand, positivistic positions had been refined to post-positivism or critical rationalism (Popper). Here, only an approximation to reality, accompanied by critical efforts of researchers to falsify hypotheses, is held to be possible, representing again the notion of an agreement process in talking about reality instead of a naive copy of reality.

Another important approach to reconcile the conflicting paradigms results from a differentiation of phases of the research process. Hans Reichenbach has worked out the difference between the first phase of defining the research question and developing hypotheses (context of discovery) and a second phase of testing hypotheses (context of justification) (cf. Hoyningen-Huene, 1987). Later on, a third phase of deriving praxis consequences from the research results (context of application) was added. In my opinion, we can follow different paradigms in different phases. Within the context of discovery and the context of application, a critical position would be important. Good research in social sciences should reflect the relevance of the research question and the possible consequences; this is an important position especially within qualitative research. But in the context of justification, a postpositivistic or moderate constructivist position would be adequate to guarantee scientific rigor.

1.2 Mixed Methods as a Solution?

In the last decades, the movement of mixed methods research has evolved as a new alternative, as a “third way” in social and behavioral science (Creswell & Plano Clark, 2010; Teddlie & Tashakkori, 2009). Models of a combination of qualitative and quantitative research approaches have been developed (Mayring, 2001; Mayring, Huber, Guertler & Kiegelmann, 2007). This movement, however, has not led to a new methodology; it puts together different steps of analysis with their different logics, mainly following a pragmatic theory of science (the methodology is adequate if it leads to the solution of the research question). Uwe Flick (1992) argues for a triangulation of qualitative and quantitative research, where each approach follows its own “method-appropriate criteria” (p. 175). But can we conduct research projects with different inherent quality criteria? Researchers looking for adequate methods are confronted with handbooks and textbooks representing the one or the other family using different criteria and sometimes including the permission to mix them up, but without a theory of integration.

Thus a methodological arbitrarism remains, best formulated in the textbook of Yin (2011), when he states,

- that the design has to be formulated at the beginning of the study **or not**;

- that you need much theory **or less**;
- that you have to plan your study **or not**;
- that the results have to be generalized **or not**.

These results are an „anything goes"-standpoint which is not satisfying.

1.3 Common Research Criteria for Qualitative and Quantitative Approaches

The best way to escape this ("postmodern") methodological arbitrarism would be formulating obligatory quality criteria valuable for quantitative as well as qualitative (as well as mixed method) research. Some efforts have been made already in the direction of defining common obliging research criteria:

- King, Keohane & Verba (1994) suggested a unified approach following a logic of inference in qualitative and quantitative approaches, but did not work out concrete criteria.

- The Keystone of Science Project (Gauch, 2003) and the National Research Council (2002) formulated criteria for qualitative projects referring to common steps of analysis (Pose significant questions that can be investigated empirically! Link research to relevant theory! Use methods that permit direct investigation of the question! Provide coherent and explicit chain of reasoning! Replicate and generalize across studies! Disclose research to encourage professional scrutiny and critique!). But this advice remained unspecific as well, because it did not provide clear methodological procedures.

- The Cochrane Qualitative Research Methods Group (Noyes, Popey, Pearson, Hannes & Booth, 2008) has listed possibilities of qualitative studies to add evidence-based reviews (Informing, enhancing, extending and supplementing reviews), but leave the quantitative-experimental gold standard.

- The American Educational Research Association AERA (2006) has formulated standards for reporting on empirical social science research in its publications, especially for qualitative projects: clear description of procedures, presentation of evidence, reasoning of interpretations and critical verification, but it does not define procedures.

On such conceptions, a valid and fruitful understanding of scientific work could be built up, which overcomes the problematic dichotomization of the qualitative versus the quantitative approach.

1.4 Qualitative Content Analysis as Mixed Methods Approach, Following Common Research Standards

The central idea of Qualitative Content Analysis is to start from the methodological basis of Quantitative Content Analysis (cf. chapter 3.1) but to conceptualize the process of assigning categories to text passages as a qualitative-interpretive act, following content-analytical rules (will be further explained in chapter 4 and 6). In this respect, the Qualitative Content Analysis is a mixed methods approach: assignment of categories to text as qualitative step, working through many text passages and analysis of frequencies of categories as quantitative step.

Furthermore, we formulate strict content-analytical rules for the whole process and for the specific steps of analysis. In this respect, our approach is dedicated to the common research criteria approach formulated above. But the Qualitative Content Analysis itself is to be understood as a data analysis technique within a rule guided research process, and this research process is bound to common (qualitative and quantitative) research standards as shown in the next chapter.

1.5 Basic Research Steps

On this basis we try to develop a step-by-step model of the research process which is valuable for qualitative and quantitative (and mixed methods) research. The model starts from traditional research processes of quantitative approaches and reformulates and expands them for qualitative approaches. Seven steps are differentiated (cf. Mayring 2001; 2012).

Step 1: Concrete research question (relevance to praxis; eventually hypotheses; formulation and explication of the researcher's standpoint)

The research questions have to be specified, expressed in a real question, not only a topic (like some qualitative projects do). Even for explorative questions, a specification is important because the results can be directly related to them (cf. step 7). Without this specification, the research process remains arbitrary. A clear research question enables one to base the research process on praxis problems and makes the research praxis relevant, which is an asset of qualitatively oriented research. Quantitative methodology on the other hand requires at this point the formulation of hypotheses in a strictly deductive thinking manner. For qualitatively oriented explorative studies, even descriptive studies, often the formulation of hypotheses is not possible, so we have to soften this requirement ("eventually formulation of hypotheses"). On the other hand, qualitative thinking often implies the conception of a researcher–subject–interaction, which means that the researcher formulates his or her standpoint in advance, and this is a form of hypotheses as well.

In chapter 8 we have introduced a recently developed open access software for Qualitative Content Analysis (QCAmap). We will give hints and explanations to this software within text blocks during the book:

> Link to QCAmap software (www.qcamap.org):
>
> This means that each Qualitative Content Analysis needs a research question as starting point, and this is implemented in the software as an obligatory text field starting the project; if there are several runs through the text, e.g. with inductive category development and deductive category application or different inductive or deductive runs, they all need specific research questions. The software program demands this from you. They can be processed parallel (cf. chapter 6.5).

Step 2: Linking research question to theory (state of the art, theoretical approach, preconceptions for interpretations)

This is a necessary step to frame research question and research results within theory, as the sum of all relevant research approaches and research results in relation to research question and subject area. Again, this is not self-evident regarding qualitative research. For example, some advocates of Grounded Theory demand not to block the open sight on the subject by theories. On the other hand, every research process is influenced by (hidden or formulated) preconceptions and only by linking research to theory a scientific progress is possible. This is especially true for interpretations. The "hermeneutical circle" (Schleiermacher) as basic procedure for interpretations means the formulation of preconceptions in advance and the stepwise modification of those preconceptions in confrontation with the material (cf. chapter 3.2).

Step 3: Definition of the research design (explorative, descriptive, relational, causal, mixed)

Following the specified research question, the adaptive research design, as the basic logic of the study, can be defined. I have shown (Mayring, 2007a; 2010) that four basic research designs can be differentiated: explorative, descriptive, correlational or causal designs. In contrast to some narrow-focused quantitative researchers, we do not believe that only causal design (experimental studies) or relational designs (correlation studies) are scientifically valuable. If explorative or descriptive studies are well formulated, they can contribute as well to important results. Furthermore, mixed designs, as just mentioned in chapter 1.2, are gaining more and more importance. Only if we accept those qualitatively oriented designs, we can apply scientific rules and rigor to them. This corresponds to the fourth claim of the National Research Council: "Provide coherent and explicit chain of reasoning!" (National Research Council, 2002).

In respect to content analysis, which is characterized by working with categories or systems of categories, the research designs have the following forms:

- Explorative design: Formulating new categories out of the material (inductive category development, cf. chapter 6.2)
- Descriptive design: Working through the texts with a deductively formulated category system (cf. chapter 6.4) and registering the occurrence of those categories, in a nominal way (category X has been found in the material) or in category frequencies.
- Relational design: Cross-tabulation of categories with person variables (e.g. comparison of category frequencies between women and men i.e. cross-tabulation category occurrences by gender), correlation (usually non-parametric) of ordinal category systems (cf. chapter 6.4)
- Causal design: A Content-analytical variable (i.e. nominal or ordinal deductive category system) within an experimental design; longitudinal analysis of category systems e.g. with biographical material. It is important to mention that causal analysis is as well possible outside a quantitative experimental design (cf. Mayring, 2007a).
- Mixed design: In chapter 6.5 several mixed content-analytical methods like typification or content structuring are described.

Step 4: Defining of the (even small) sample or material and the sampling strategy

Even if qualitatively oriented studies often work with small samples, with single case studies, they have to describe and give arguments for the sample size and sampling strategy. The sample, as the empirical basis of the research project, can consist of documents (different files, web-pages), persons (interviews e.g.), situations (field notes) or broader entities (e.g. groups, cities). In any case, a sampling strategy has to be developed. Random sampling is only one of those strategies (even sometimes relevant in Qualitative Content Analysis, e.g. newspaper analysis); cluster samples, stratified samples, grouped in respect of theoretical considerations, or stepwise explorative sampling in the form of “Theoretical Sampling” (Glaser & Strauss, 1967) are possible procedures. Convenient samples or ad-hoc-samples, i.e. the researcher taking what he gets without any argumentation, should be avoided. If it is the only solution, then the possibilities of generalization of the results are widely restricted.

Link to QCAmap software (www.qcamap.org):

Within the software package the “cases” of the sample consist in documents. For each research question those documents (interview transcripts of different persons, field notes, files …) the relevant documents have to be divided into different text files and converted in Unicode (txt).

Step 5: Methods of data collection and analysis, pilot tested

Clear methodological procedures in data collection and data analysis are basic within quantitative and qualitative approaches. A good argumentation for a specific technique often consists of a comparison to an alternative technique. So projects working with Qualitative Content Analysis have to give arguments why they do not use another text analysis procedure, e.g. quantitative content analysis or Grounded Theory Coding (cf. for an overview chapter 2). Within quantitative approaches usually standardized procedures, for example test instruments, are used. On the other hand, within qualitative approaches the instruments (interview agenda) are developed for the specific study and they have to be pilot tested.

In Qualitative Content Analysis the category systems are developed inductively out of the concrete material or deductively put together individually for the specific study. Therefore, those elements have to be pilot tested as well for gaining methodological strength. This is possibly very easy because the textual material can be processed several times. In the step-by-step models of inductive and deductive categorization (cf. chapters 6.3 and 6.5) a pilot study element is always formulated to test and modify the category systems.

Link to QCAmap software (www.qcamap.org):

After the first coding, the software program automatically gives a hint, that the category system needs a pilot test phase. You can decide, whether it is too early or you can proceed with this pilot phase following the step-by-step model. If the category system or the central content-analytical rules (category definitions, level of abstraction, coding agenda) are changed as a result of the pilot test, the material has to be coded again from the beginning.

Step 6: Processing of the study, presentation of results in respect to the research question

So we have seen, that any changes of the instruments, and of course changes of the research question have the consequence of a new process of the step-by-step model. Qualitative researchers often characterize the research process as cyclic (in contrast to the linear quantitative research process, moving from research question to results). We consider the possibilities of changing instruments and even the research question within the project as sometimes important, but then we put the same rigor to the new instruments or research question.

At the end of processing the study it is important for quantitative and for qualitative studies to present the results in a broad descriptive sense and in the more specific sense of answering the research question.

Step 7: Discussion in respect to quality criteria

A critical discussion of the own research results seems to be crucial for a scientific approach. The classical criteria, deriving from the test theory (objectivity, reliability and validity) cannot be simply transferred to qualitative approaches (cf. Steinke, 2000). But an introduction of totally different criteria seems to be problematic as well. A position, influenced by a constructivist theory of science, that qualitative and quantitative approaches, each following their own quality criteria, can be combined by triangulation (e.g. Flick, 2007) is not compatible with our intention of a unified scientific process. I think, validity in a broader sense is usually less of a problem within qualitative approaches, because they seek to be subject centered, close to everyday life (naturalistic perspective, field research), especially when the research process remains theory driven (construct validity). In qualitative research, efforts have to be made to enhance reliability in a broader sense. Within Qualitative Content Analysis, the rule guided procedures can strengthen this criterion. Objectivity, defined as total independence of the research results from the researcher, is held to be difficult within qualitative approaches. But on the other side, they discuss the interaction researcher–subject and strengthen objectivity in a broader sense.

> Link to QCAmap software (www.qcamap.org):
>
> For Content Analysis in particular, several specific quality criteria have been developed like inter-coder and intra-coder agreement, which will be discussed in chapter 6. Both criteria are implemented in the software program: on the project page an agreement button opens the possibility to share the project with a second coder or coding process and to compare the results (cf. chapter 7).

An overview of theses seven steps, which make up a general step-by-step model of the research process, is given in the following figure (for specific content-analytical step-by-step models see chapter 4.6 and the example in chapter 5).

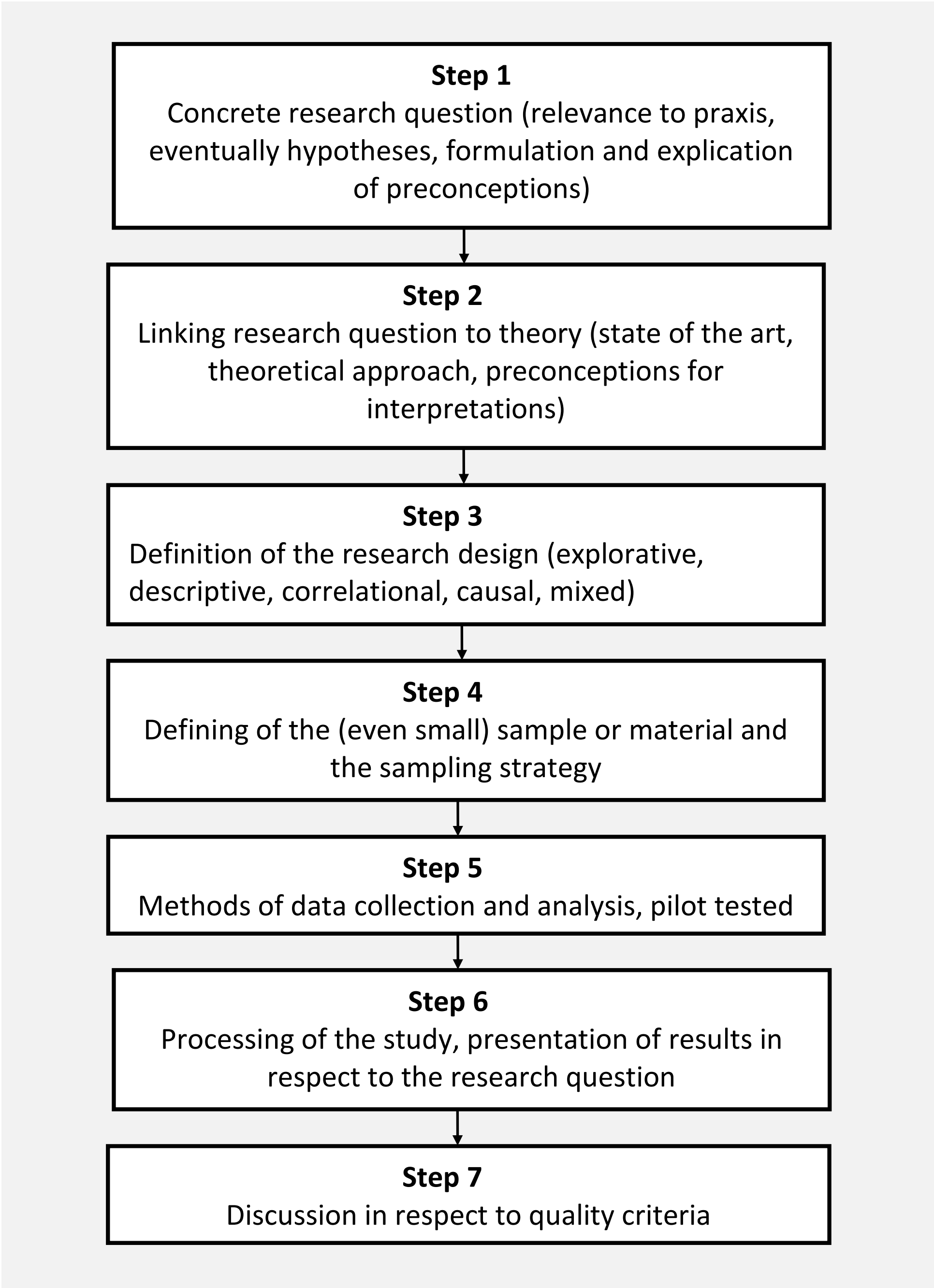


Figure 1: Step-by-step model for the research process

Such a step-by-step model can be a point of reference for quantitative, qualitative and of course for mixed methods research. And in this way perhaps the unfruitful "science war" in social science methodology can be overcome.

2. Overview on Approaches to Text Analysis in Social Sciences

We have just mentioned that working with Qualitative Content Analysis needs an argumentation in respect of its adequateness. For this reason it is useful to look at alternative text analysis procedures in social sciences. Perhaps we can differentiate between three traditions that modern text analysis techniques are coming from:

For hermeneutic approaches, coming from a background of humanities ("Geisteswissenschaften") background, the text has to be interpreted by the formulation of the own preconceptions (hermeneutical circle); the intensions of the text author have to be found out and an additional explaining text has to be formulated. The tradition originates from theology (interpretation of bible texts) and jurisprudence (interpretation of law texts). In the figure below (Fig. 1) we have listed six modern hermeneutical approaches:

- Objective hermeneutics has been developed in Germany by sociologist Klaus Oevermann (Reichertz, 2000) with the aim of drawing inferences to objective social structures behind the text. An elaborated technique of sequence analysis has been formulated even if the interpreter has broad degrees of freedom in his interpretation (interpretation as art).
- Grounded Theory (Corbin & Strauss, 1998) describes a procedure of coding textual materials (e.g. a more inductive open coding process and a more deductive axial coding process) and defining the codes with memos. The aim is to come to a concrete theoretical model by means of an explorative process.
- Psychoanalytical text interpretation (Koenig, 2004) was developed to draw inferences from the text to a deep structure of defended contents. By logical analysis, fractures or inconsistencies in the text are discovered which can be a sign for a defense mechanism in the author.
- Phenomenological analysis has been developed in psychology (Giorgi, 2009) originating from philosophy (Husserl, Heidegger). The phenomenon is analyzed through variation and reduced to its core concept.
- Biographical analysis (Miller, 2005) interprets open-ended textual materials on individual life courses. If those approaches analyze the formal structure of the biographical text as narration (narrative structure), they take in linguistic consideration, which is expressed in figure 2 by a link.

Linguistic considerations have inspired several approaches especially within cultural studies under the label of the Discourse Analysis (Gee & Handford, 2013). Usually, the first step of those approaches follows a linguistic criterion (in metaphor analysis the identification of metaphors in the text, in conversational analysis the reconstruction of the interaction process) and then interprets the result in a more hermeneutical way. Discourse Analysis in a narrower sense embeds the textual material in the discursive situation in which it is located. Text mining procedures include more explorative strategies of quantitative text analysis, which sometimes includes content-analytical procedures.

Content Analysis (cf. chapter 3.1) has been developed within communication science to analyze huge textual corpuses (e.g. newspapers) in a first quantitative way. There are connections to linguistics (text mining). In the second half of the 20th century qualitative approaches, like ours, have been formulated.

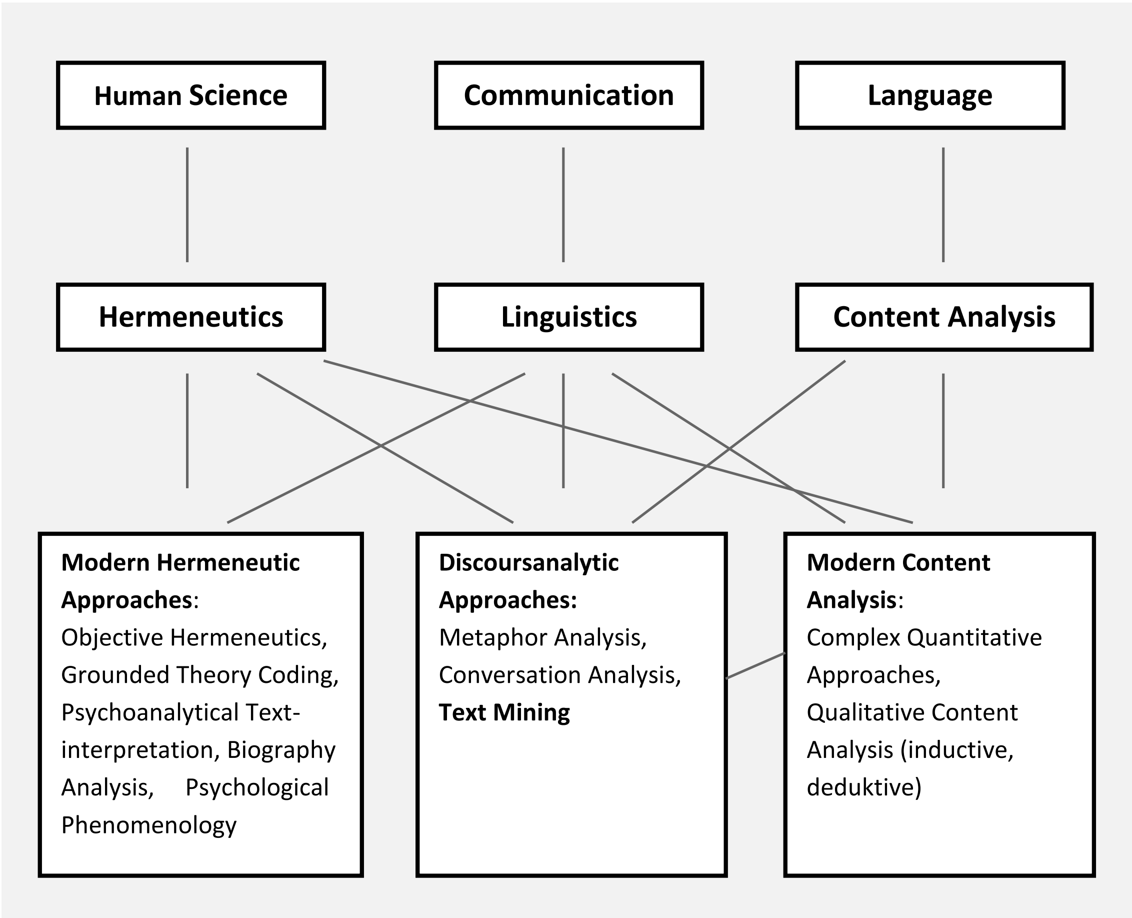


Figure 2: Approaches to Social Science Text Analysis

Working with one of those text analytical procedures does not mean that the scientist has to come from the underlying discipline, but we have to take into account the background. Like in quantitative data analysis, we have to choose the adequate statistical operation, we have to determine the preferred text analysis technique within qualitative approaches and to give arguments for this decision.

The advantages and limitations of Qualitative Content Analysis are discussed in chapter 9.

3. Theoretical Background for a Qualitative Content Analysis

The theoretical foundation for the development of procedures for a qualitative content analysis can be found in different areas:

3.1 Communication Science: Quantitative Content Analysis

It is possible to distinguish between three phases in the development of content-analytical techniques and approaches (cf. on this point Berelson, 1952; Merten, 1983; Franzosi, 2004):

3.1.1 Preliminary Phase

Content analysis certainly has a relatively short history, but it may as well have a long past. For attempts to analyse communication material systematically can be traced back through the centuries. In the 7th century, for instance, word-frequency analyses of Old Testament texts were carried out (Yule, 1944). During the doctrinal controversy between Lutherans and Pietists in the 18th century their texts were subjected to a comparative content analysis. It was shown that certain key concepts (God, Kingdom of Heaven) occurred with the same frequency and that therefore no fundamental deviation from orthodoxy on the part of the Pietists could be proven (cf. Dovring, 1954).
Around the turn to the 20th century we find less quantitative approaches in the analysis of language material as well, like the dream analyses of Sigmund Freud.
The first systematic newspaper analysis, one of the main fields of early content analysis, dates from as early as 1893 (Speed, 1893). Here the news articles were assigned to certain thematic categories and compared across different papers (Tribune, World, Times, Sun).

Table 2: Newspaper analysis of Speed, 1893 (Merten, 1983, p.36)

Subject	Tribune 1881	Tribune 1893	World 1881	World 1893	Times 1881	Times 1893	Sun 1881	Sun 1893
Editorial	5.00	5.00	4.75	4.00	6.00	5.00	4.00	4.00
Religious	2.00	0.00	0.75	0.00	1.00	0.00	0.50	1.00
Scientific	1.00	0.75	0.00	2.00	1.00	0.00	0.00	2.50
Political	3.00	3.75	0.00	10.50	1.00	4.00	1.00	3.50
Literary	15.00	5.00	1.00	2.00	18.00	12.00	5.75	6.00
Gossip	1.00	23.00	1.00	63.50	.50	16.75	2.00	13.00
Scandals	0.00	1.50	0.00	1.50	1.00	2.50	0.00	2.00
Sporting	1.00	6.50	2.50	16.00	3.00	10.00	0.50	17.50
Fiction	0.00	7.00	1.50	6.50	1.00	1.50	0.00	11.50
Historical	2.50	2.50	2.75	4.00	2.50	1.50	4.25	14.00
Music and Drama	2.50	4.00	1.50	11.00	4.00	7.00	0.00	3.50
Crimes and Criminals	0.00	0.50	0.00	6.00	0.00	1.00	0.00	0.00
Art	1.00	1.00	3.00	3.00	2.00	0.00	0.25	1.25

The illustration shows an index (deviation from average according to article and photo sizes) for the treatment of individual topics in the four newspapers, compared on two randomly selected publication dates. It demonstrates that religious, scientific and literary topics are losing ground, whereas gossip, scandal and crime are increasing.

3.1.2 Consolidation Phase

On the basis of such studies, content analysis consolidated itself into a standard instrument of empirical social research. In the initial decades of the last century, content analysis was developed first of all in publishing and journalism as a systematic method of analyzing news articles. A decisive contribution was made in this respect by the Columbia University School of Journalism (cf. Willey, 1926). In the late thirties the method received great impetus. Responsible for this were the following factors:

- Mass media such as radio and newspapers were becoming increasingly important. Analyzing them was part of the attempt to discover “public opinion”. It was in this connection that the Bureau of Applied Social Research at Columbia University was set up under the chairmanship of Paul F. Lazarsfeld.
- During Second World War the Experimental Division for the Study of Wartime Communications had been instituted by Congress to assess precision propaganda under the chairmanship of Harold D. Lasswell.
- The Department of Justice commissioned content analyses for domestic intelligence purposes.
- Commercial contractors (e.g. the press, General Motors) also discovered that it was a method they could use.

Against this background the first monograph was written on content analysis by Berelson (1952), who developed it as an objective, systematic and quantitative analysis of the manifest content of communication.

3.1.3 Fine Developments and Interdisciplinary Expansion

Following this development, content analysis was also taken up by other disciplines (e.g. psychology, sociology, educational science, historical science, fine arts studies). The method received new impetus through the conference on content analysis held by the Committee on Linguistics and Psychology of the Social Sciences Research Council in 1955 at Allerton House, University of Illinois, Monticello (Allerton House Conference) (cf. Pool, 1959). It was established on this occasion that:

- not only the summarizing of verbal material (description) was important, but also the conclusion (inference) to be drawn from the material on the circumstances of its origin and effects;
- in the material not only symbol frequencies but also symbol connections are measurable (contingency analyses);
- qualitative procedures can also be useful: A. L. George criticized quantitative content analysis and demanded that it be complemented by a "non-frequency approach" (cf. George, 1959);

- the problem of the meaning of symbols must also be discussed; one cannot simply start from the lexical meaning of terms but should also take into account their context, their circumstances of origin and the intentions behind them (cf. Mahl, 1959).

A good ten years later the second important conference on content analysis was held at the University of Pennsylvania´s Annenberg School of Communication in Philadelphia (Annenberg School Conference of 1966). The most important further developments here were as follows (cf. Gerbner, Holsti, Krippendorff, Paisley & Stone, 1969):

- an attempt was made to analyze the analytical procedure itself more precisely (the "content-analytical situation", cf. Krippendorff 1969a).
- the demand was made that the theoretical model of communication on which the analysis is founded (cf. Ch. 4.4) should be explained (Krippendorff 1969b).
- compromise positions emerged in the controversy between qualitative and quantitative analysis (Holsti and Gerbner in Gerbner et al., 1969).
- quantification techniques were made more accurate. Extensive computer programs were developed (cf. Gerbner et al. 1969, Part IV).

3.1.4 The Present-day Situation: “Discontent” Analysis?

Discussion of content analysis as an instrument of the communication theory did not essentially pass beyond this point (cf. Krippendorff, 1980). The method was also applied outside the United States (cf. e.g. Lagerberg, 1975, d'Unrug, 1974). It was used in Germany, for instance, from the end of the 1950s onwards (cf. Silbermann, 1967; Rust, 1981; Merten, 1983). Quantitatively oriented content analysis became the standard instrument of the empirical communication science.

However, one can say that at this point the methodology discussion has reached a point of stagnation. An increasing number of critical voices described the technique as inadequate and unable to fulfil requirements. The joke about “discontent analysis” was to be heard with increasing frequency. Koch, Witte & Witte (1974), for example, tested six fairly recent journalistic content analyses from German-speaking countries according to customary standards of quality. In their opinion content analysis gets a bad report: “If conclusions are drawn on the basis of the work reviewed here, then it must be stated that up to now no one has succeeded in developing a handy instrument for describing and analyzing news publications with the help of content analysis” (Koch, Witte & Witte, 1974, p. 83, translation P.M.).

Manfred Ruehl also denied that content analysis has a chance of achieving “social-scientific status capable of gaining general acceptability” (Ruehl, 1976, p.377). It achieves only superficial polish through quantitative techniques, and has pushed the problem of sense and meaning to one side, he argues. “The results of content analysis remain highly pseudo- and parascientific, as long as content analysts do not know how to equip their scientific criteria better for methodological testing” (Ruehl, 1976, p. 376/377).

The fact, that the quantification approach and orientation to manifest content tends to sidestep the problem of what language symbols actually mean, was reason enough, also for Ingunde Fuehlau, to declare that content analysis is a failure. “This is why content analysis, if pursued strictly according to its own tenets, must inevitably lead to distorted results. If the method was stringently applied - which actually is almost never really the case - it must either produce irrelevant descriptions of the subject - albeit in a very ‘objective manner’ - or on the other hand meaningful descriptions of communication content, to which, however, if judged according to its own criteria, it can only assign a highly subjective value. In either case, therefore, it fails as a method” (Fuehlau, 1978, p. 15/16, cf. also Fuehlau, 1982).

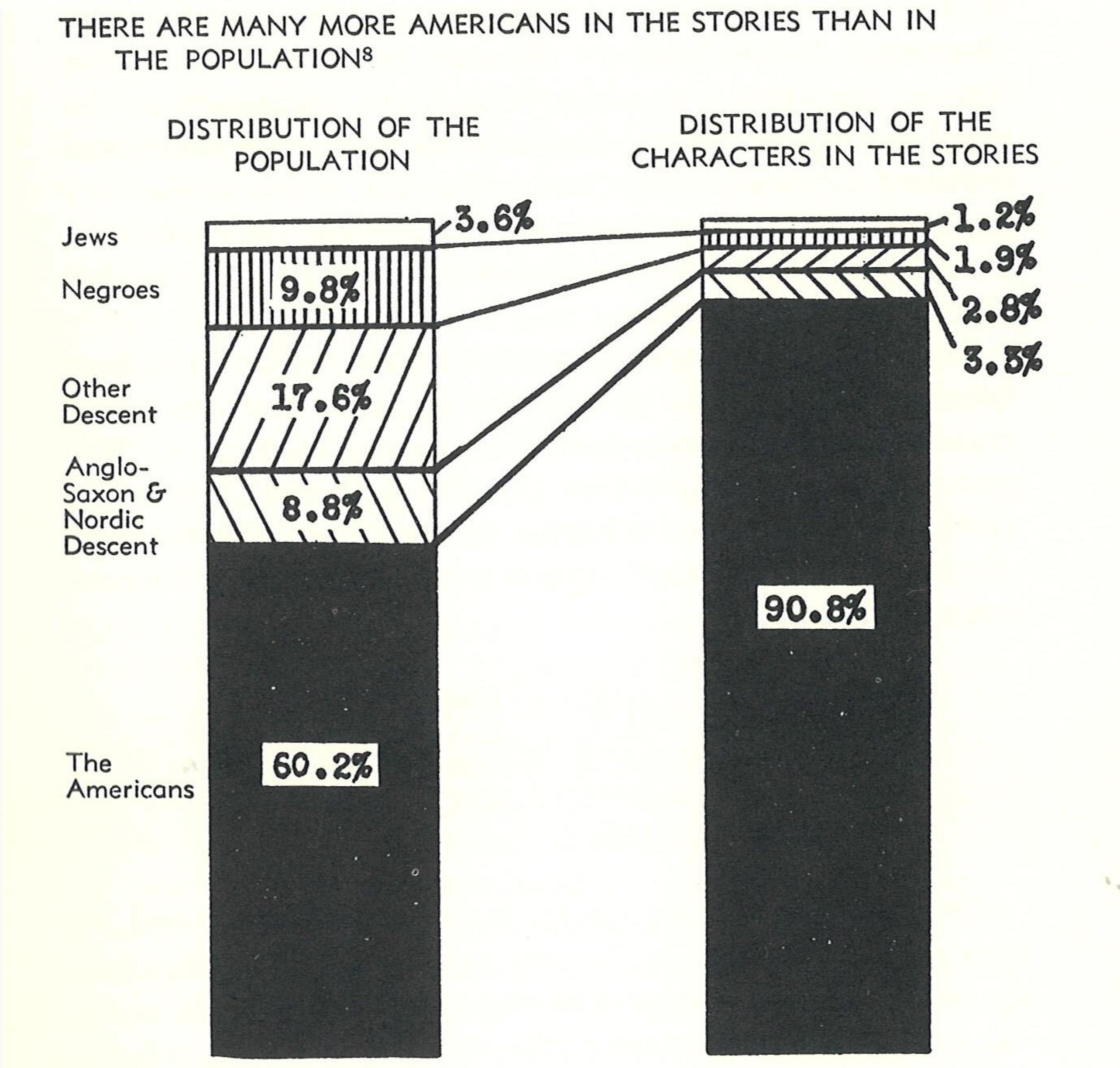
Certainly, communication sciences have made positive attempts to overcome the shortcomings of the classical content analysis. Hitherto, however, these have remained on the level of theoretical programmes and have been unable to suggest concrete techniques (e.g. Kracauer, 1972). One thrust in this direction is Holger Rust’s conception of qualitative content analysis (Rust 1980a, 1980b, 1981). He conceives of qualitative content analysis as a qualification, as “classifying and determining the contours of the object under examination within its context, delineating it relative to other objects and generally characterizing its inner consistency” (Rust 1981, p. 196). In other words, it includes everything for which any form of quantification prepares the groundwork. Qualitative content analysis must take the structure and meaning of the material to be analyzed (i.e. the text) as its starting point. The construction of a text, according to Rust, is therefore the basis of the method.

1. Any text entails the stylizing of information.
2. In stylizing certain information the text gives relevance to certain meaning relationships.
3. Through this semantic units are built up, the size of which must be determined and varied in order to disclose inner principles of construction and external relations.
4. The subordinate units of text are marked and delineated.
5. The relationship of the subordinate units to other areas of content or the behavior behind it is characterized.
6. These relationships can be expressed through certain patterns, which can vary in size.
7. The divisions between subordinate semantic units can be overcome again on the basis of the particular cultural background involved.
8. For the recipient certain subordinated semantic fields are recognizable as stylizations of his or her everyday life (cf. Rust, 1980a, p. 12/23).

“Qualitative analysis therefore pursues a double strategy: it forces the object of analysis to reveal its structure in a de-totalizing approach which inquires into the relationship between individual aspects and general appearance, but does this with the aim of achieving a conscious re-totalization, so as not to lose sight of the overall social core content of every statement” (Rust 1980a, p. 21). Rust himself calls this a theoretical outline, and admits that concrete procedures are missing entirely (Rust, 1981, p. 201). This is characteristic of the situation in which qualitative content analysis finds itself.
Other approaches had been developed in the area of content analysis, like codebook analysis (Neuendorf, 2002). Here a non-automatically, manual (interpretative?) coding is used, following a codebook with explicit code definitions and sometimes examples. It seems to be similar to deductive category assignment (cf. chapter 6.5), but in codebooks it is not described and worked out systematically and theoretically founded (cf. chapter 9 for further content analytical approaches).

3.1.5 Basic Techniques of Quantitative Content Analyses

It is **frequency analyses** and techniques derived from them that should be mentioned primarily here. The simplest method of a content-analytical procedure is to count certain elements in the material and compare them in their frequency with the occurrence of other elements. Here is a simple example: In 1946 B. Berleson and P. Salter (Berelson, 1952) carried out an inquiry into the ethnic origins of the main figures in American magazine stories, comparing the percentage distribution with the actual ethnic distribution in American society:



8 The foreigners in the stories are omitted here. The data are not strictly comparable but provide the best comparison which could be secured from available information. The population data, taken from the World's Almanac for 1944, are from the 1940 census (except for the figures for Jews, which come from the Jewish Statistical Bureau). The data on hyphenates represent the foreign-born and the native-born of foreign and mixed parentage. (In this respect, there is a bias against the distribution of minority characters in the stories, who are of any generation.) The figures for the Americans were secured by subtracting the total for the other groups from the total population.

Figure 3: Content Analysis "American Majorities and Minorities" from Berelson 1952, p. 51

Of special importance here is the use of comprehensive category systems (so-called "dictionaries"), which are supposed to include all aspects of a text and form the basis for a computer count of language material. The General Inquirer (Stone, Dunphy, Smith and Ogilvie, 1966) seems to have been the first attempt in this direction. Dictionaries now exist, for instance, for psychologically relevant issues (e.g. Harvard Psychological Dictionary), the latest editions of which can be conveniently used on a PC (cf. Weber, 1990; http://www.wjh.harvard.edu/~inquirer/). Figure 4 shows the encoding of two sentences from speeches of candidates for the US Presidency in 1980 (left-hand column) and the categories assigned on a word-for-word basis (right-hand column).

Table 3: Computer-aided quantitative content analysis of two sentences with Harvard IV Psychological Dictionary; Weber, 1990, p. 33

```
   Word              Categories
SENTENCE7**DOCUMENT1**IDENTIFICATION AD1980
   THE              ARTICLE
   EFFECT#1         ABSTRACT CAUSAL PSV
   ON               SPACE
   OUR              AFFILIATION OUR
   ECONOMY          DOCTRINE ECONOMIC
   MUST#1           OUGHT
   BE#1             BE
   ONE#2            INDEF OTHER
   WHICH            INDEF INT RLTV1
   ENCOURAGE#1S     INTERRELATEL AFFILIATION PSTV ACTV
   JOB              MEANS ECONOMIC
   FORMATION        MEANS STRNG
   AND              CONJ1
   BUSINESS#1       DOCTRINE ECONOMIC
   GROWTH           STRNG INCR PSV
***START NEWX DOCUMENT..
SENTENCE8**DOCUMENT2**IDENTIFICATION AR1980
   TAX#1ES          MEANS POLIT ECONOMIC
SENTENCE9**DOCUMENT2**IDENTIFICATION AR1980
   ELSEWHERE        SPACE
   IN               SPACE
   THIS#1           DEM DEM1
   PLATFORM#1       DOCTRINE POLITICAL
   WE               PLRLP OUR
   DISCUSS          PSTV COMFORM
   THE              ARTICLE
   BENEFIT#35       GOAL PSTV STRNG
   FOR              CONJ CONJ2
   SOCIETY          COLL POLITICAL
   AS#1             CONJ2 CAUSAL
   A                ARTICLE
   WHOLE#2          QUAN STRNG OVRST
   OF               PREP
   REDUCED          DECR STRNG
   TAXTATION,       MEANS POLIT ECONOMIC
   PARTICULAR#4LY   OVRST
   IN               SPACE
   TERM#1S          COM COMFORM
   OF               PREP
   ECONOMIC         POLIT DOCTRINE ECONOMIC
   GROWTH.          STRNG INCR PSV
```

On this basis frequencies are computed and analyzed statistically. Of course the dictionary must also be able to recognize different grammatical forms of a word within the context of a sentence. This, however, can cause problems:

- multiplicity of meaning (e.g. "madly" in the colloquial meaning, say, of "very"; or "madly" as pertaining to psychological disturbance);
- the nuances and connotations conferred on terms by the context;
- contextual modification of meaning (for instance in the case of "no anxiety", "little anxiety" and "a lot of anxiety", "anxiety" will be counted once in each case);
- the contextual relationship of the term counted (e.g. with "I am afraid of X" or "X is afraid of me", "afraid" is counted once in each case);
- the problem of pro-forms (e.g. with "I didn't notice any of that" the computer does not know what "of that" refers to);
- dialect expressions (which occur in interview scripts regularly) need a great deal of re-working.

And several more problems could be added to the list. Attempts have in fact been made to check and control contextual influences of this kind (KWIC Keyword-in-Context-Program, cf. Weber, 1990). For this purpose a list of the points of appearance of a category, that is, the category in its different contexts is drawn up for each concept or term counted. Figure 5 shows a section from it on the category "rights" in the above-mentioned example (speeches of candidates for US presidency).

Table 4: Key-word-in-context list for the category ‘rights’; Weber, 1990, p. 45

1980 Reagen Republican Platform

YOUNG PEOPLE WANT THE OPPORTUNITY TO EXERCISE THE	RIGHTS AND RESPONSIBILITIES OF ADULTS. THE REPUBLICAN PA	AR 1980	372
ACTERIZED BY THE HIGHEST REGARD FOR PROTECTING THE	RIGHTS OF LAW-ABIDING CITIZENS, AND IS CONSISTENT WITH T	AR 1980	1004
OF THEIR SCHOOL SYSTEMS. WE WILL RESPECT THE	RIGHTS OF STATE AND LOCAL AUTHORITIES IN THE MANAGEMENT	AR 1980	333
RIGHTS AND THE HELSINKI AGREEMENTS WHICH GUARANTEE	RIGHTS SUCH AS THE FREE INTERCHANGE OF INFORMATION AND T	AR 1980	1391
UALLY AND STEADFASILY COMMITED TO THE EQUALITY OF	RIGHTS FOR ALL CITIZENS, REGARDLESS OF RACE. AS THE PART	AR 1980	206
S ISSUES, IS ULTIMATELY CONCERNED WITH EQUALITY OF	RIGHTS UNDER THE LAW. THERE CAN BE NO DOUBT THAT THE QUE	AR 1980	284
SE WHO SUPPORT OR OPPOSE RATIFICATION OF THE EQUAL	RIGHTS AMENDMENT. WE ACKNOWLEDGE THE LEGITIMATE EFFORTS	AR 1980	227
SSION ARE IN THE COURTS. RATIFICATION OF THE EQUAL	RIGHTS AMENDMENT IS NOW IN THE HANDS OF STATE LEGISLATUR	AR 1980	232
REAFFIRM OUR PARY’S HISTORIC COMMITMENT TO EQUAL	RIGHTS AND EQUALITY FOR WOMEN. WE	AR 1980	228
XEMPTION FROM THE MILITARY DRAFT. WE SUPPORT EQUAL	RIGHTS AND EQUAL OPPORTUNITIES FOR WOMEN, WITHOUT TAKING	AR 1980	229
ON POLICY MUST BE BASED ON THE PRIMACY OF PARENTAL	RIGHTS AND RESPONSIBILITY.	AR 1980	322
N’S COMMITMENT TO DEFENT THEM. INDIVIDUAL	FEDERAL EDUCATI RIGHTS AND SOCIETAL VALUES ARE	AR 1980	152
MULTIRACIAL SOCIETY WITH GUARANTEES OF INDIVIDUAL	ONLY AS STRONG AS A NATIO RIGHTS IS POSSIBLE AND CAN WORK.	AR 1980	1557
VE ECONOMIC SECURITY. HISPANICS SEEK ONLY THE FULL	REPUBLICANS BELIEVE THA RIGHTS OF CITIZENSHIP -- IN	AR 1980	213
UNITIES FOR WOMEN, WITHOUT TAKING AWAY TRADITIONAL	EDUCATION, IN LAW ENFORCEMEN RIGHTS OF WOMEN SUCH AS	AR 1980	229
ING STRONG, EFFECTIVE ENFORGEMENT OF FEDERAL CIVIL	EXEMPTION FROM THE MILITARY DRAF RIGHTS STATUTES,	AR 1980	209
CARE IS DEREGULATION AND AN EMPHASIS UPON CONSUMER	ESPECIALLY THOSE DE DURING THE NEXT FOU RIGHTS AND PATIENT	AR 1980	350
IMPLEMENT THE UNITED NATIONS DEGLARATION ON HUMAN	CHOICE. THE PRESCRIPTION FOR GOOD HEA RIGHTS AND THE	AR 1980	1391
THEIR EMIGRATION IS A FUNDAMENTAL AFFRONT TO HUMAN	HELSINKI AGREEMENTS WHICH GUARANTEE RIGHT RIGHTS AND THE	AR 1980	1394
BEEN DURING THE CARTER ADMINISTRATION. HUMAN	U.N THE DECLINE IN EXIT VISAS TO SOVIET J RIGHTS IN THE	AR 1980	1398
N’S RHETORIC. THE MOST FLAGRANT OFFENDERS OF HUMAN	SOVIET UNION WILL NO BE IGNORED AS IT HAS RIGHTS INCLUDING	AR 1980	1072
NS LINKED TO IST UNDIFFERENTIATES CHARGES OF HUMAN	THE SOVIET UNION, VIETNAM, AND CUBA HAV RIGHTS VIOLATIONS.	AR 1980	1473
	YET, THE CARTER ADMINISTRATION’S POLI		

This, however, only makes it possible to recognize the problem, not to remove it. In any case, lists such as this are difficult to process with large quantities of text.

The basic procedure for such frequency analyses, also regarded as a model for more complex analyses, is as follows:

- formulation of issue or problem;
- determination of the material sample;
- establishment of a category system (dependent upon the issue concerned), i.e. determination of which text elements are to be checked for frequency;
- definition of the categories, possibly with examples;
- determination of analysis units, i.e. decision as to
  - what the minimum component of text is that can fall under the heading of a category (recording unit),
  - what the maximum text component is (context unit) and
  - the sequence in which text components are to be encoded (unit of classification); such components can be syllables, words, sentences, paragraphs, etc.;
- coding, i.e. working through the material with the help of the category system in order to record the occurrence of categories;
- computation, i.e. establishing and comparing frequencies;
- description and interpretation of the results.

One example of a more complex frequency analysis is the Gottschalk-Gleser Speech Content Analysis for the measurement of affective states (anxiety, aggression) (Gottschalk & Gleser, 1969), which has also been adapted for the German language (Schoefer, 1980).

The next group of established quantitative techniques to be mentioned are **valence and intensity analyses**. Generally speaking these are content-analytical procedures which accord a value to certain textual components on an assessment scale of two or more gradations. The general procedure can be described as follows:

- formulation of issue or problem;
- determination of the material sample;
- establishment and definition of the variables to be examined;
- determination of the scale values (features per variable), with valence analyses bipolar (e.g. plus - minus), with intensity analyses multi-graded (e.g. very strong - strong - medium - less strong - null);
- definition and possible addition of examples for the scale values of the variables (variables and scale values together constitute the category system of these analysis types);
- determination of analysis units (recording unit, context unit, unit of classification);
- coding, i.e. scaling of the assessment units according to the category system;
- computation, i.e. establishment and comparison of frequencies of scaled assessments, possibly further statistical processing;
- description and interpretation of the results.

Valence and intensity analyses may be constructed very simply, e.g. when the leader articles of several daily newspapers are compared with regard to how far they support the policies of the governing party or those of the opposition. Three examples of more complex forms can be mentioned here: the symbol analysis, the evaluative assertion analysis (Osgood, Saporta & Nunally, 1956) and the value analysis (White, 1944).

This brings us to the third group of tested techniques of content analysis: **contingency analyses**. The development of such techniques goes back mostly to Charles Osgood (Osgood, 1959). The objective here is to establish whether particular text elements (e.g. central concepts) occur with particular frequency in the same context, whether they are connected with one another in any way in the text, i.e. whether they are contingent. The intention is that by discovering many such contingencies one may extract from the material a structure of text elements associated with one another. Quite generally the procedure can be defined as follows:

- formulation of the issue;
- determination of the material sample;
- establishment and definition of the text components whose contingency is to be examined (i.e. drawing up of a category system);
- determining the units of analysis (recording unit, context unit, unit of classification);
- definition of contingency, i.e. establishing rules as to what counts as a contingency;
- coding, i.e. working through the material with the aid of the category system;
- examination of common occurrence of the categories, establishment of the contingencies;
- collation and interpretation of the contingencies.

Examples of this are the classical contingency analysis of Osgood's (1959), discourse analysis (Harris, 1952), semantic field analysis (Weymann, 1973) and the association structure analysis (Lisch, 1979).

3.2 Human Sciences: Hermeneutics

Hermeneutical approaches generally are an important source for the development of the qualitative research methodology. In some respect the Qualitative Content Analysis as well refers to it.

Hermeneutical approaches have the longest tradition of text analysis (cf. Bruns, 1992). In Greek mythology the messenger of the gods was Hermes; his duty was to translate, to interpret, to communicate the intentions of Zeus, which is the basic idea of hermeneutics. The later fields of hermeneutics were theology, jurisdiction, history and philology. In those cases the aim is to give interpretations of central texts (bible, laws, historical documents, literature), to comment those texts, always in the sense of understanding the real intentions of the text authors.

Several philosophers have outlined the central procedures of hermeneutical text understanding. Mathias Flacius Illyricus (1520-1575), theologist, a scholar of Martin Luther and Philipp Melanchton, elaborated the idea of understanding single text passages on the background of the overall text and its context. Friedrich Schleiermacher (1768-1834), philosopher, defined hermeneutics as the understanding of meaningful reality (not only texts) as an art (“Kunstlehre”) more than a formal method. Friedrich Ast (1778-1841), classical philologist, formulated the hermeneutical circle as central procedure of text understanding. That means that the interpreter has to formulate his or her preconception, preknowledge (“Vorverstaendnis”) of the topics of the text. Then he or she reads the text and modifies the preconceptions. (In some respect this procedure has similarities with hypotheses guidedness of quantitative research.) Later on the term “hermeneutical spiral” was preferred, because the interaction between preconceptions and text interpretations show a dialectical development and not only a circle. Figure 4 visualizes this spiral process:

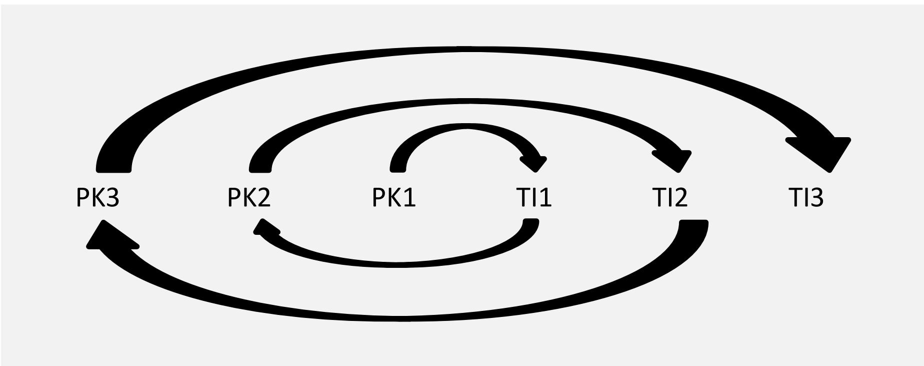


Figure 4: The hermeneutical spiral (cf. Danner, 1979) (PK: preknowledge; TI: Text interpretation)

Wilhelm Dilthey (1833-1911) defined hermeneutics as an artistic method of understanding (“Kunstlehre des Verstehens”) and conceptualized it as the basis for human sciences like mathematics are the basis for natural sciences. But he did not formulate a dichotomy: On the fundament of more descriptive hermeneutical understanding a second step of scientific explanations and correlations can be conducted. That seems to be a very modern concept, nowadays discussed under the approach of mixed methods.

In the meantime several researchers elaborated the concept of hermeneutics (e.g. Heidegger, Gadamer, Betti, Habermas). Coreth (1969) is outlining on this background four central ideas of the hermeneutical process of understanding:

- Horizon structure: specific text passages can only be understood on the basis of the whole text and its context as background.
- Circle structure: texts can only be understood as relation between preknowledge and preconceptions of the interpreter and the text itself.
- Dialog structure: text understanding is embedded in an interaction process between text author and text interpreter.
- Subject-object structure: In the text real life objects are mentioned and again there is an interaction process between the subjects involved (author, interpreter, audiences) and those text objects.

In the previous chapter we just mentioned that nowadays there are several approaches of text analysis on an explicit hermeneutical background (e.g. Objective Hermeneutics). What does this mean for Qualitative Content Analysis?

We would say that the hermeneutical approach to text analysis is important. It reminds us that text understanding is not an automatic process of counting manifest text elements (like in Quantitative Content Analysis). On the other hand qualitative Content Analysis includes systematic quantitative steps of analysis. I like to demonstrate the hermeneutical elements within Qualitative Content Analysis with an example from our work (Mayring, 2002b):

This example comes from a study on psycho-social consequences of unemployment (Mayring, Koenig, Hurst & Birk, 2000). Fifty teachers becoming unemployed in consequence of the German unification after 1990 took part in open-ended interviews. The material was transcribed and analyzed by qualitative content analysis. One step of analysis was to apply categories in a deductive way to the text. So we tried to appraise the degree of stress of the interviewed persons, working with three deductive categories: no stress, little stress and high stress.

The Coding Agenda contained definitions and coding rules like the ones listened in figure 5:

Category	Definitions	Coding Rules
no stress	no negative aspects; only subjective unimportant stresses whole situation positive	coping efforts not necessary
little stress	single negative factors for the subject; pos. and negative aspects in the situation	coping possibilities seem to be clear
high stress	overall negative situation; some severe bad aspects, depressed, insecure	no coping possibilities are seen

Figure 5: Part of the coding guideline for stress categories

The purpose of those content-analytical rules is to make the process of category application as controlled as possible. Let us now look at one of the interviews:

CASE X

I: Is it a stressing situation for you now?

A: ...(reflecting) ... Well, that's a difficult question. Until now, I have to say, I'm not through with this, because it had been so disappointing. You got your next job, you had to fight for it, and now I'm employed for a probationary period at the youth hostel, I hope to get the job in June, and to bring in my experiences as teacher, I think it's a big challenge.
.... But sometimes I'm feeling depressed, for example if you don't know how to manage a situation in the new job. But I hope things will come to a good end.

After the first sentence of the answer we think the teacher is highly stressed, because he is troubled with the situation, the situation is unclear, is disappointing. In the next sentence he tells us, that he has managed the situation perfectly. He speaks about a new challenging job, about hope. No unemployment stress would be the right coding. But then he tells us something about feelings of depression and the impossibility to cope with the situation, a sign for a high stress coding. A clear decision, what category would be adequate is only possible on the background of the whole interview and is not an automatic process of coding rule application.

A second text example from another interview out of this study may underline this point:

CASE Y

I: Well how is the situation at the moment, is it stressing?
B: Yes, well I think that one is not able to cope with this, that they simply push you aside.
I: And what is the central problem for you?
B: Well, the injustice. That they took things into account for their decision, which are not right.
I: Are there any positive aspects in your situation now?
B: Well, I would say, I'm not bad in my new job selling contracts for the building society, I got used to it very well, I'm one of the best. That's always with me, to be better than the others. But, well, it's a job I haven't chosen by myself. And if you are looking at the employed teachers, this is hard. But on the other hand I'm glad that I don't have to work in this educational system any more.

Here again the decision for a category swings from sentence to sentence. He shows us a hopeless situation with no possibilities to cope. But he as well has found a new job and is very motivated in it. Perhaps as a form of defense he tells us that he is glad to be out of his former teacher job. Here we understand that we need to have background material to understand his situation (development of the educational system after the German reunion). Again we don't see a simple automatic coding process. Even if the coding agenda is more elaborated, containing further coding rules and text examples for clarification, the coding remains a complex act of interpretation.

On this background we try to discuss the role of a researcher within the content-analytical work. The two poles of orientation are:

- being only part of the research instrument, applying content-analytical rules in a mechanical, automatic way, trying to be constant, observable, intersubjectively understandable and able to be checked by inter coder reliability tests;
- or being a free interpreter of the material, having content-analytical steps and rules only as orientation, establishing a subjective relation to the material.

We tried to argue that qualitative content analysis remains interpretation. The central step of relying categories and parts of the text material is not an automatic technique but a reflective act of interpreting meanings in the text. So the procedures of quantitative (e.g. computerized) content analysis are fundamentally different. The content analyst has to put all his competencies, pre-knowledge and empathic abilities into the process of analysis. But he has to do this within the framework of content-analytical rules.

Link to QCAmap software (www.qcamap.org):

Coding the texts remains a decision process of the researcher. In one part of the screen the textual material is presented, relevant text passages have to be marked with the cursor and related to categories. On the same screen all relevant content-analytical rules are displayed to support the decision. The text can be scrolled to have an overall impression of the material in respect to the category. The codings can be changed if the researcher revises his or her decisions.

3.3 Linguistics: The Structure of Language and Text

If we try to develop procedures of text analysis, we have to understand what text is and what language is. The scientific discipline covering this area is linguistics (Akmajian, Demers, Farmer & Harnish, 2010; Schulte-Sasse & Werner, 1977). And indeed we have just mentioned some text analysis procedures, which are based directly on linguistic concepts (metaphor analysis, conversation analysis, discourse analysis, see chapter 2).

Semiotics, as a part of linguistics, is defined as the analysis "of the exchange of meanings of acting or communicating individuals" (Schulte-Sasse & Werner, 1977, p. 49, transl. P.M.), and this is very relevant for the text analysis. Semiotics differentiates between

- the used language signs,
- the people using those signs,
- the objects to which the signs are related,
- the ideas of the objects in the mind of the users.

So text analysis can follow very different questions:

- How is the text constructed out of different signs (*syntactics*)?
- What are the meanings of the signs, how could they be interpreted (*semantics*)?
- What is the relation between signs and users (*pragmatics*)?
- What is the relation between signs and objects (*sigmatics*)?

In chapter 3.1 we have defined content analysis as a systematic procedure of assignment of categories to portions of text. The question which now occurs is: what could be text portion, sentences, phrases, words? Within the procedures of content analysis (as well of Qualitative Content Analysis) the analyst is forced to define those parts in advance, called content-analytical units (cf. chapter 4.4). This definition of content-analytical units determines how subtle or rough the text analysis will be. The definition depends on the research question and the quantity of material.

So what are the possibilities for defining those units? Linguistics differentiates the following elements:

- **Seme** is the smallest meaning component of texts (Greimas, 1983; Schulte-Sasse & Werner, 1977). Structural semantics hold that specific language terms can bear several meaning aspects. Seme means the smallest unit. So terms for seating furniture can be understood as combination of different semes:

  S1: furniture
  S2: only to sit
  S3: with backrest
  S4: with armrest
  S5: with legs
  S6: hard material
  S7: cushioned
  S8: only for one person

  A sofa would be a combination of S1, S3, S4 and S7, a stool a combination of S1, S2, S5, S6. But sofa can contain other semes like coziness or bourgeois.

- **Phoneme** is the smallest hearable segment of language, a sound or tone.

- A **syllable** is the phonological (sound elements to be heard) unit of words. Words can have one or more syllables.

- **Words** are the basic elements of texts, which have a lexical meaning. Words can have different meanings in respect to their text context ("blue" as a color or a mood).

- **Phrases** are groups of words without finite verbs, which have a syntactic (grammatical) connection.

- A **Paraphrase** is the content of a phrase without any decorative or filler words, it is the core meaning of the phrase. The semantic content is equivalent to the phrase, but is expressed in a short form.

- **Clauses** are parts of sentences with syntactic (grammatical) connection and verbs.

- **Sentences** are speech units, which are complete and relatively independent in respect to grammar, content and intonation.

- A **proposition** is, similar to a paraphrase, the content of a sentence, the logical statement, independent from the language form.

- **Paragraphs** are (usually) two or more consecutive sentences which have a common meaning or theme. In interview transcripts paragraphs are made between questions and answers.

- **Text documents** are paragraphs belonging together, usually from one communication source or situation of emergence.

Link to QCAmap software (www.qcamap.org):

The software forces you to define content-analytical units (if not defined you cannot code your texts). You have to define the coding unit, the context unit, and the recording unit (see chapter 4.4). For that you can use those linguistic terms.
For summarizing content analysis (cf. chapter 6.1) the concept of paraphrases would be helpful.

Linguistics can help us to develop procedures of text analysis in another way: for the procedure of explication of unclear text passages we have to define what determines the meaning of a part of text. From linguistics we get two answers:

- The lexical and grammatical meaning,
- The context meaning.

Lexical and grammatical meaning can easily be discovered by formal analysis of the text. Context meanings are more difficult. We have to define, what context means. Van Dijk (1999; 2007) has worked out a linguistic theory of context. For him every talk and every text is situated and therefore needs a context analysis. “It is the way participants *understand* and *represent* the social situation that influences discourse structures” (Van Dijk, 2007, p. 4). The context gives a frame of reference. He differentiates two models of context:

- **The micro context**: that is the specific situation (time, location, the speaking (writing) person, his or her identity, aims, personal knowledge and his actions and plans).
- **The macro context:** that is allocation in society, the relevant reference groups and group actions and goals, the institutional and cultural background.

We derive from this differentiation two forms of explicating content analysis, narrow and broad context analysis, and use those descriptions for the development of content-analytical rules (cf. chapter 6.3).

Link to QCAmap software (www.qcamap.org):

To implement explicating qualitative content analysis (narrow and broad context analysis) within the QCAmap-software is a plan for the future (because it is not used so often like inductive category development and deductive category assignment).

3.4 Psychology of Text Processing

Another research field seems to provide knowledge for developing text analysis techniques: the psychology of text processing (Ballstaedt, Mandl, Schnotz & Tergan, 1981; Mandl, 1981). This is an area within educational psychology, which analyses everyday processes of students working with texts. Researchers try to observe persons dealing with texts in educational or everyday environments. One promising method of data collection in this context is "thinking aloud". The person in front of the text formulates and speaks out all the cognitive processes (perceptions, appraisals, thoughts), which are going on in himself or herself.

Text processing is understood as interaction between reader and text, as an active construction of meaning structures by the reader. His or her preknowledge and interests have a selective and organizing function within this process. Text understanding is guided by cognitive schemata. "A schema is an active organizing unit of knowledge, which based on experiences brings together different concepts of objects, events and actions within one complex of knowledge" (Schnotz, Ballstaedt & Mandl, 1981, p. 113, transl. P.M.).

The psychology of text processing now differentiates between an ascending (starting with the text) and a descending (starting with a schema) direction of text understanding. Ballstaed, Mandl, Sachnotz & Tergan (1981) have demonstrated this in the following figure:

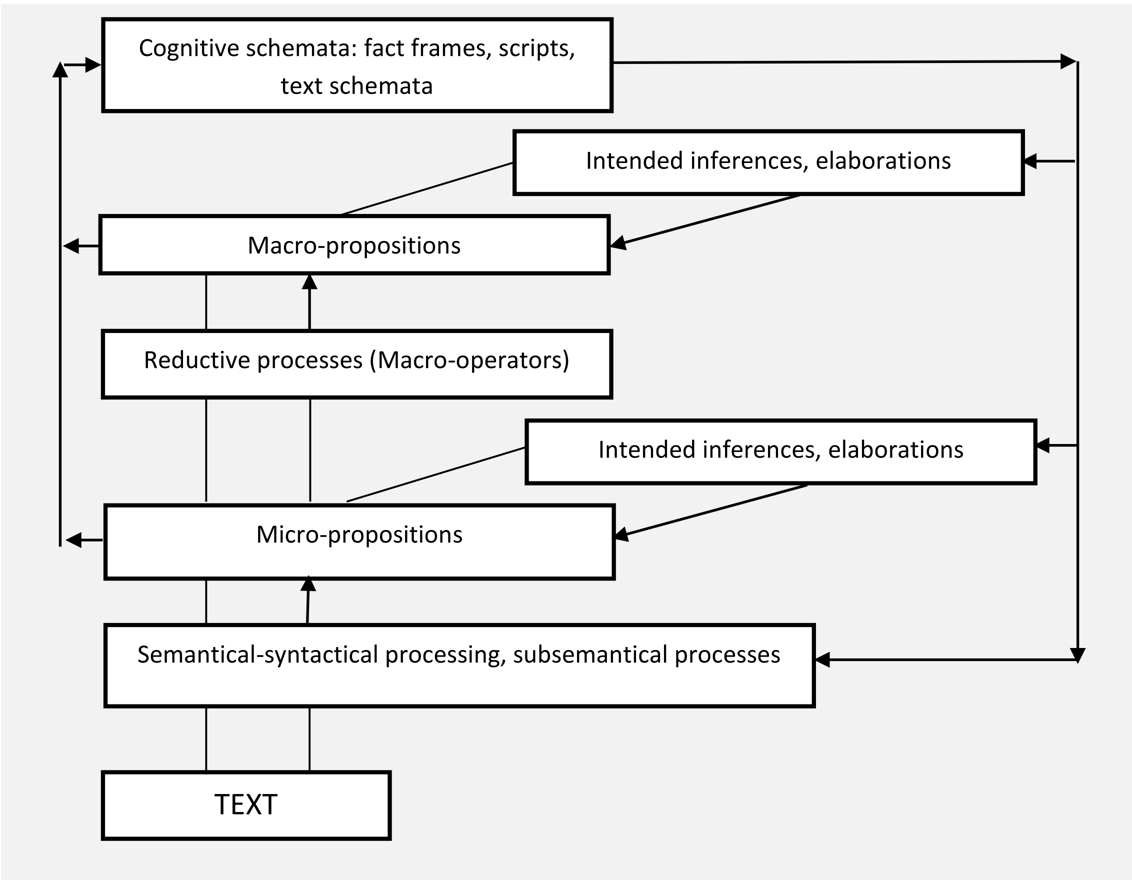


Figure 6: Model of the processes of text understanding (Ballstaedt et al., 1981, p. 83)

The text (at the bottom of the model) first is realized visually (subsemantical processes), characters, words etc. are identified in their meanings and relationships (semantic-syntactic processing) to build up a network of meaning units (micro-propositions). Here the model borrows concepts from linguistics (cf. chapter 3.3). At this point already preknowledge and preconcepts, cognitive schemata, are used:

The reader adds to the text own experiences in the sense of elaboration or inferences. The next steps, so the theory says, and empirical studies have shown, are reductive: the text is summarized to a smaller network of meaning units (Macro-propositions). This macro-structuring again is described in linguistics (VanDijk, 1980). The studies of everyday processes of learners summarizing texts could differentiate five different strategies of reduction:

1. Leaving out

Propositions of a text could be left out, if they are not necessary for the understanding of other propositions and if they are not the result of Macro-proposition. Ballstaedt et al. 1981 (p. 70ff) gave an example:

> "Because the world, following a well-known slogan, became smaller through airplanes, satellites, and television…"

The hint to the well-known slogan is not necessary for the understanding of the whole text and can be left out.

> *"Because the world became smaller through airplanes, satellites, and television…"*

2. Generalization

Related propositions in the same context could be summarized by a more general, more abstract paraphrase with a superordinate meaning. It serves as macro-proposition. This could be related as well to parts of propositions, predicates and arguments. Here again the example:

> "Because the world, following a well-known slogan, became smaller through airplanes, satellites, and television…"

could be summarized by generalization to:

> *"Because the world, following a well-known slogan, became smaller through means of transportation and media…"*

3. Construction

In a series of propositions belonging to a comprehensive, more global fact a new proposition can be constructed, which formulates the common overwhelming meaning. Here again an example from Ballstedt et al., 1981):

> "He took the matches, lit the pipe and puffed the smoke into the air."

Could be summarized by construction into

> *"He smoked."*

4. Integration

The process is similar to construction, but here the summarizing proposition is already found within the text.

> "He took the matches, lit the pipe and smoked."

Could be summarized by integration into:

> *"He smoked."*

5. Selection

In a broader context, a central proposition is chosen from the text basis, because its content seems so important that it could not be left out. In this case, the original proposition and the summarizing proposition are identical. The reader finds within a text a sentence which bears the central idea (normally he underlines the sentence) and selects it.

If the reader, using those five reductive operators, arrives at macro-propositions he again links them with inferences and elaborations from his or her pre-knowledge (cf. Fig. 6).

> Link to QCAmap software (www.qcamap.org):
>
> The psychology of text processing especially those reductive operators (leaving out, generalization, construction, integration, selection) can be used to formulate content-analytical rules for summarizing.

3.5 General Psychology: Theories of Categorization

The next important research field originates from general psychology. We have learned in the introduction (chapter 1), that the central elements of all forms of content analysis are the categories. They are the instruments with which the text is worked through. They can be inductively developed out of the material or deductively crystallized from theory and then assigned to parts of the text.

But what are categories? General psychology analyses the processes of learning and memory, of mental representation of the world (Muesseler & Prinz, 2002). Concepts and categories are central terms in those cognitive processes. A basic procedure of knowledge building is to put things we experience together into classes of things. Concepts are mental representations of classes of things, “concepts are the glue that holds our mental world together” (Murphy, 2002, p. 1). Categories are the classes themselves.

It was Aristotle (384a-322a), the developer of the first comprehensive system of sciences, who put the process of categorization in the center. Every science has to construct basic categories and main categories and to order the objects of its research area into those categories. So we arrive at a descriptive theory of the discipline. The classical viewpoint on categories (Murphy, 2002; Waldmann, 2002) is, that there are defining criteria for each concept. A triangle is defined as closed geometrical form with three straight sides including three angles with a sum of 180°. But another possibility of defining categories would be to list some examples. Not only general psychology was interested in those rules of defining categories as a central component of human knowledge. Developmental psychology (e.g. Jean Piaget) analyzed how children are learning categories, which would be an important part of speech development cognitive development, respectively. Following these lines of research we nowadays differentiate between three theories of categorization (Murphy, 2002):

- The **definitional theory**, coming from the classical view of categories, lists necessary and sufficient conditions for belonging to the category. On the basis of this explicit definition the classification of objects is possible.
  Example: A tree is a plant with a central wooden trunk, lateral branches with leaves or needles.

  There are some critical points within the definitional theory: the limits between categories are often unclear, especial with natural categories (Is a chicken a bird?). Categories may overlap. The rules often are so complex that the language user does not know them.

- The **prototype theory** holds that we have in mind typical exemplars of each category. We compare the objects that we observe with those prototypes, and if they are similar we can categorize them.
  Example: A typical tree would be (at least for a Bavarian) a fir.

  This explains that some exemplars of a category are more or less typical, that there are maybe blurred limits. But as well this is the problem of the approach: only the core of the category is defined.

- This leads to the third approach: the **decision bound theory**. The categories are defined by their differences to neighbor categories, the language user knows the limits within a set of similar categories.
  Example: A tree has in contrast to a bush only one trunk, is usually higher and lives longer.

  But this approach was criticized because it cannot explain what sort of mental representation stands behind a category.

If each of those categorization theories has disadvantages, perhaps the best possibility to define concepts is to use all three approaches for definition. And in fact some researchers have developed an approach of multiple systems in categorization (Waldmann, 2002). The language user switches in his or her mental representation between definitional and demarcation rules and typical examples of categories. The most precise definition of categories would be to use all three approaches.

> Link to QCAmap software (www.qcamap.org):
>
> For deductive category assignment the exact definition of the categories is crucial. We use all three approaches for all categories (definitions, anchor examples and coding rules) and put them together in a coding guideline. It is developed before coding using theoretical arguments (especially the definitions) and completed (anchor examples, additional coding rules) within the pilot phase.

4. Basics of Qualitative Content Analysis

4.1 Basic Principles and Definition

The basic approach of qualitative content analysis is to retain the strengths of quantitative content analysis and against this background to develop techniques of systematic, qualitatively oriented text analysis. This will be explained more closely in the following.

4.1.1 Embedding of the material within the communicative context

A particular advantage of content-analytical procedures as compared with other approaches to text analysis is the fact that it has a firm basis in the communicative sciences. The material is always understood as relating to a particular context of communication. The interpreter must specify, to which part of the communication process he wishes to relate his conclusions from the material analysis. This content-analytical particularity should be retained at all costs for qualitative content analysis because many quantitative content analyses have neglected this point. The text is thus always interpreted within its context, i.e. the material is examined with regard to its origin and effect. A complex model in this connection will be introduced in the next chapter.

4.1.2 Systematic, rule-bound procedure

Preserving the systematic procedure of content analysis is one of the main concerns of the methods suggested here. Systematic procedure in this connection means first and foremost: orientation towards rules of text analysis laid down in advance. This is seen at several points. The establishing of a concrete procedural model of analysis is of central importance. Content analysis is not a standardized instrument that always remains the same; it must be fitted to suit the particular object or material in question and constructed especially for the issue at hand. This is defined in advance in a procedural model (examples of such models will very frequently be found during perusal of this book), which defines the individual steps of analysis and stipulates their order. But it is also continually necessary to establish additional rules. Such bodies of rules are featured below. It is an axiom precisely of content analysis, in contrast to "free analysis", that every analytical step and every decision in the evaluation process should be based on a systematic and tested rule. Finally, the systematic quality of content analysis is reflected also in its method of "dissection". The definition of content-analytical units (recording units, context units, coding units, cf. chapter 4.5) should on principle be retained also in qualitative analysis. Concretely, this entails deciding in advance how the material is to be approached, which parts are to be analyzed in what sequence, what conditions must be obtained in order for an encoding to be carried out. In the process of inductive category formation it can be useful to keep such content-analytical units very open-ended. Despite this, however, the process here also is characterized by dissection of the material carried out progressively from one passage to the next. Certainly, it is precisely this last point, which has frequently been criticized by proponents of the qualitative approach. Latent structures of meaning cannot be revealed in this way, they say. One answer to this, in the case of such an analytical objective, is to define the units in an accordingly broad fashion. Nevertheless, it is important that such units are theoretically well founded, in order to allow other analysts to access the logic and method of the analysis. The system should be described in such a way that another interpreter may carry out the analysis in a similar way.

4.1.3 Categories in the focus of analysis

The category system is the central point in quantitative content analysis. Even with qualitative analysis, however, an attempt should be made to concretize the objectives of the analysis in category form. The category system constitutes the central instrument of analysis. It also contributes to the intersubjectivity of the procedure, helping to make it possible for others to reconstruct or repeat the analysis. In this connection qualitative content analysis will have to pay particular attention to category construction and substantiation. However, precious little help is given in this respect by standard works on content analysis. Krippendorff thus writes: "How categories are defined ...is an art. Little is written about it." (Krippendorff, 1980, p. 76). That of course is unsatisfactory. It is precisely the methods described in this work, which may be of further assistance in this regard. On this point also, qualitative proponents make the objection that orientation to categories entails an analytically dissecting methodology which impedes synthetic comprehension of the material. In answer to this it can be said that qualitative content analysis also provides methods which accord prominence to synthetic category construction, i.e. where the category system actually constitutes the findings of the analysis. On the other hand, working with a category system is an important contribution to the comparability of findings and the evaluation of analysis reliability.

4.1.4 Object reference in place of formal techniques

On the other hand the methods of qualitative content analysis should not simply be techniques to be employed anywhere and everywhere. The alliance with the individual object of analysis is an especially important concern. This is seen in the fact that the procedures discussed here are oriented to the way language material is ordinarily experienced and dealt with in everyday life. The three base techniques of summarizing, explication and structuring (cf. chapter 6) are based on this and the rules for those basic procedures stem from an analysis of everyday handling of texts (cf. chapter 3.3, 3.4 and 3.5). This clearly demonstrates that it is the object of analysis which is paramount. The methods are not intended to be conceived of as techniques which can be blindly and automatically transferred from one object to the other. The appropriateness of method must be demonstrated with regard to the particular material in each individual case. This is why the methods suggested here must always be adapted to suit the individual study.

4.1.5 Testing specific instruments via pilot studies

Regarded from the viewpoint of traditional quantitatively oriented scientific understanding, this last point could be objected to on the grounds that it provides no guarantee of methodological comparability. Qualitatively oriented content analysis, however, deliberately forgoes the use of fully standardized instruments precisely because it places relations with the individual object above all else. This is why methods must first be tested in a pilot study. This applies equally to the fundamental method and the specific category system. In the procedural models in chapter 6 these steps are already included through the presence of reverse loops. What is important in this is that the trial runs are also documented in the research report. Here the inter-subjective testability is again of central importance, too.

4.1.6 Theory-guided character of the analysis

It must now have become clear that qualitative content analysis is not a rigidly delineated technique, but a process in which new decisions regarding basic procedure and individual stages of analysis constantly have to be made. What are such decisions based upon? In qualitatively oriented research it is repeatedly stressed that theoretical arguments must be used. Technical fuzziness is compensated for by theoretical stringency. This applies above all to the explication of the particular issue, but it also concerns detailed analyses. Theory-guidedness means that in all procedural decisions systematic reference is made to the latest research on the particular subject and on comparable subject fields. In qualitative content analysis, content-related arguments should always be given preference over procedural arguments; validity is regarded more highly than reliability.

4.1.7 Integrating quantitative steps of analysis

As was already emphasized in the last chapters, efforts are made to combine qualitative and quantitative methods. Putting it more exactly, the chief task is to determine those points in the analytical process at which quantitative measures can be sensibly brought in. Reasons for their use should then be carefully explained and the results should be analyzed in detail.

Quantitative steps of analysis will always gain particular importance when generalization of the results is required. In case study procedures it is important to show that a certain case recurs in similar form with particular frequency. But within content-analytical category systems, registration of how often a category occurs may give added weight to its meaning and importance as well. Of course, this must be given adequate justification in the respective case. A precisely based qualitative assignment of categories to a certain material (e.g. through the structuring method, cf. chapter 6.5) can also be supplemented by more complex statistical evaluation techniques, as far as these are appropriate to the purpose of analysis and suited to the object involved. Especially attractive in this connection are the computer programs developed in the last few years as a support for qualitative analysis (cf. chapter 6). Here qualitative and quantitative steps of analysis have been made generally available in the simplest possible way, which lends particular support to integrative methodological conceptions.

4.1.8 Quality criteria

It is precisely because here the harsh methodological standards of quantitative content analysis have been softened and applied more flexibly in some respects, that the assessment of results according to quality criteria such as objectivity, reliability and validity is especially important even in qualitative content analysis (cf. on this point Ch.7). For content analysis it is inter-coder reliability which is of particular significance. Several content analysts work on the same material independently from one another and their findings are compared. In general this should also be attempted with qualitative content analysis, although negative findings do not necessarily have to lead to the immediate abandoning of the analysis. Here the main point, again, is to understand and interpret unreliabilities. Such a search for sources of error is especially important during the pilot phase, as it can lead to the instruments of analysis being modified. That is to say, it can lead to inquiry into arguments for reliability and validity while the process of analysis is actually going on, instead of leaving this exclusively to a single assessment at the close of the analysis.

4.2 Materials for Qualitative Content Analysis – What Could be Analyzed?

Content Analysis is a method of data analysis. Sometimes, e.g. within mass media research contexts (cf. chapter 3.1), it is labeled as data collection method, because it extracts material (as sample) out of a huge amount of texts (e.g. newspapers). But this seems misleading for us. The step of sampling material from text corpora (in the context of social sciences we would call this a document analysis design) is done before content analysis. As lined out earlier (chapter 1.4) a sampling theory would be necessary, or at least arguments for the selection of material. But what would be possible material for Qualitative Content Analysis?

When we have finished the process of data collection, as possible material for answering the research question, there are two classes of results: numerical data (frequencies of test or questionnaire values, tallies in standardized observation studies, measurements) or texts. It is a pity that textbooks on data analysis mostly only deal with the analysis of numerical data (which means statistical analysis) and leave out text analysis. But texts are occurring so often within social science contexts, like:

- Interview transcripts: There are different forms of interviews like narrative interview, biographical interview, deep interview, focus interview, semi-structured interviews, which are all leading to transcripts.
- Focus groups: It is a more and more favored data collection method to hold moderated group interviews. The discussions are recorded and transcribed.
- Materials from open questionnaires: Many questionnaire studies contain at least some open questions, which are leading to text material.
- Observational studies which are not fully standardized (in the sense of fixed checklists or tallies) produce protocols. Especially in field studies it is important to write field notes. All of this produces text material.
- Document analysis as research design can deal with a broad range of texts: newspapers or other mass media products, files, protocols, documentations in institutions, web pages and so on.
- Secondary analysis is a more and more interesting research approach, because scientific institutions are building up databases of study materials like texts, which are free for further text analysis.

For all studies which are producing their text material themselves (interview, focus group, open questionnaire or observation) it is important to decide for transcription rules. There are different models (cf. Howitt, 2010, chapter 6), handling dialect, verbal and nonverbal characteristics through special signs (see chapter 4.3). It is crucial to decide for a system of transcription and to employ it constantly. The text analysis can only refer to the transcripts, and transcripts are never complete representations of their raw material.

Link to QCAmap software (www.qcamap.org):

To import the text material into the software it is necessary to have a text file in Unicode, an international digital standard format. Following an ISO-norm, signs from different alphabets like Arab, Greek, Kirill, Hebrew, Thai, Japanese, Chinese as well as mathematical, economic and technical special characters can be read. Only bold face and underlines are ignored. Use capitalization or spacing for accentuations.

In some cases a transcription would be too much time and resource consuming, especially if the material is clear, less ambiguous, and the research question needs no deep interpretation. Then the analysis could be done directly from the tape-recorded material. The techniques of Qualitative Content Analysis could be applied. Even video material could be analyzed using Qualitative Content Analysis (cf. Mayring, Glaeser-Zikuda & Ziegelbauer, 2005). In those cases the video material is treated as text, because the categories have to be defined as text. A direct coding of video material without referring to language is, at the moment, not possible.

Link to QCAmap software (www.qcamap.org):

The use of the QCAmap software would of course not be possible in that case, because it needs text material. Maybe in the future we will develop possibilities for implementing audio or video files.

4.3 Transcription Systems

The transformation of spoken language (in an interview or a focus group) into text needs transcription rules. The interview transcript almost always implies a loss of information, a focus on only some aspects of the spoken language. Usually the content of the language is of main interest, but there are possibilities to enrich the text with additional aspects. A transcription system is a set of exact rules how spoken language is transformed into written text. I have put the following transcription systems into order depending on how much information is preserved (and in consequence how time consuming the transcription process will be) (cf. Edwards, 2002; Howitt, 2010, chapter 3.6).

- **Selective protocol**: This is an economic procedure for transcription. The researcher defines those parts of the (audio recorded) interview, which are relevant for the research question. Interviews often contain extensive introductory parts, motivating the person or explaining the research question, excurses which are important for maintaining a good climate and the compliance of the interviewee. But those parts sometimes are not necessary for the text interpretation. Or the interview has an open, narrative character and the researcher is only interested in specific topics. The researcher formulates a clear selection criterion and the transcription regards only those passages.

- **Comprehensive protocol**: If the material is not too ambiguous, not too open to interpretations, and if we are interested only in the content, a comprehensive protocol might be sufficient. The material is on hand in textual (documents) or audio-recorded (interview) form. The researcher reads or hears the language, stops in regular periods and sums up the main content writing it down or speaking it into a microphone. In the last case the use of an automatic speech recognition program could be useful for the transcription. It has to be trained for the own voice; because of this necessity of training the adoption for ordinary interviews is not recommendable. Of course the researcher has to be trained for the summary procedure.

- **Clean read or smooth verbatim transcript**: The transcription is done word for word, but all utterances like uhms or ahs, decorating words like, right, you know, yeah are left out. A coherent text, simple to understand but representing the original wording and grammatical structure is produced. Short cut articulation and dialect are translated into standard language (c’mon = come on).

- **Pure verbatim protocol**: The transcription is done word for word including every utterance from the audio file. Dialect formulations, fillers, articulation are maintained. The transcript now is very near to the natural language, but reading it is not easy, sometimes (e.g. slang) needs some practice.

- **International Phonetic Alphabet (IAP):** If we want to preserve as much as possible the coloration in oral language (like dialects) in transcripts we can use the International Phonetic Alphabet (see http://www.langsci.ucl.ac.uk/ipa/) with special characters, usually used in foreign language dictionaries, to indicate the pronunciation. Some of those special characters are (sounds of a):

a	open, short
ɐ	close light
ɑ	dark open
ɒ	round
æ	open light
ɑ̃	nasal
ʌ	dark closed

The problem of this system of transcription is, that you need a special set of characters and that the text is not easy to read. But sometimes it makes sense to use this technique.

- **Protocol with special characters**: This technique is usually used for interviews in qualitative research. There is a set of signs for describing nonverbal aspects of the natural language. Above all every characteristic like laughter, crying, low voice is notated. There are different systems in different countries (languages). In German speaking countries the GAT system of transcription (Selting, Auer, Barden & Bergmann, 1998) is widely used. Here are some examples of symbols and meanings:

acCENT	capitals for accentuations
ac!CENT!	strong accentuation
?	pitch rise
;	lower pitch
< p >	quiet speech (piano)
((laughter))	special language events
( )	not understandable passage
(.) (….)	small or long pause
: :::	small or long lengthening

For the English language the Jefferson transcript system (Jefferson, 2004) is widely used. It uses for example ↑for pitch rise (“absol↑utely”) and ↓for lower pitch (“absolutel↓y”) and ° for quieter speech (“she had °died°”), other signs are used similar to GAT.

Link to QCAmap software (www.qcamap.org):

All those special characters, including the signs in the International Phonetic Alphabet, are kept when the text is transferred in Unicode-txt-format, which is necessary for the software. Only bold, cursive and underlining are ignored.

- **Protocol with comment column:** This maybe most extensive form of protocol allows the transcriber to use a special column for all special perceptions besides the text. This procedure sometimes is used for the transcription of focus group discussions. Along with the discussion moderator a second researcher is present in the groups and writes down an observation protocol, which then is united with the text transcript.

It becomes clear that a certain system of transcription has to be defined and argued. It is important to give the exact rules at hand to the transcribing person. The decision for one of those systems depends on the research question, the characteristics of the language, and the theoretical background of the analysis. For a psychoanalytical text analysis for example a word-by-word transcription including nonverbal aspects seems to be very important. Other procedures do not demand this elaborateness. The decision for one system might be a matter of resources (time and money) as well.

4.4 Content-Analytical Context Model

When the base material has been described in this way, the next step is to ask what one would like to find out from it. Without a specific line of inquiry or established direction of analysis any content analysis would be unthinkable. The text cannot be interpreted "off the cuff", as it were. Determining the line of inquiry can be conceived of as a two-stage operation:

*** Direction and goal of the analysis**

Language material allows statements to be made in a variety of directions. One can describe, for example, the subject matter treated in the text, one can discover something about the author of the text, or establish the effect of the text on the target reader. This is something that must be decided in advance. What is helpful in this respect is to perceive the text as part of a communication chain, and to integrate it into a content-analytical communication model. An approach is given by Lasswell's formula on the analysis of communication: "Who says what, in what way, to whom and with what effect?" A simple communication model on this basis would be the following (Lagerberg, 1975):

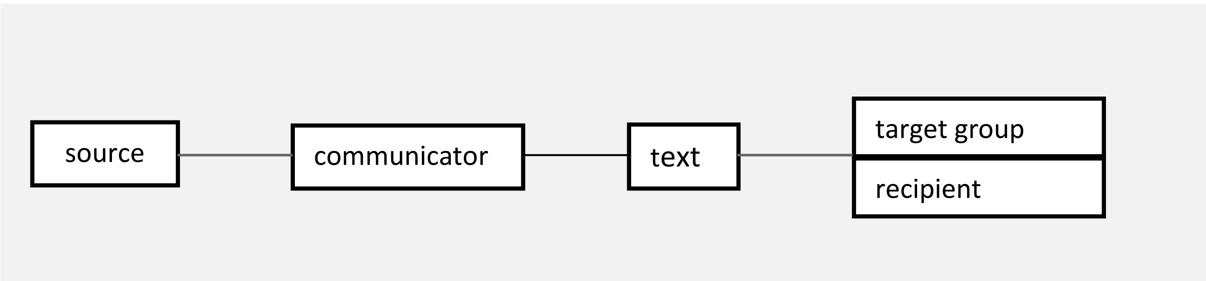


Figure 7: Simple content- analytical communication model (Lagerberg, 1975)

On the basis of what has been discussed in the preceding chapters, however (cf. chapter 5.2: Defining the base material), this model must be extended (Fig. 8).

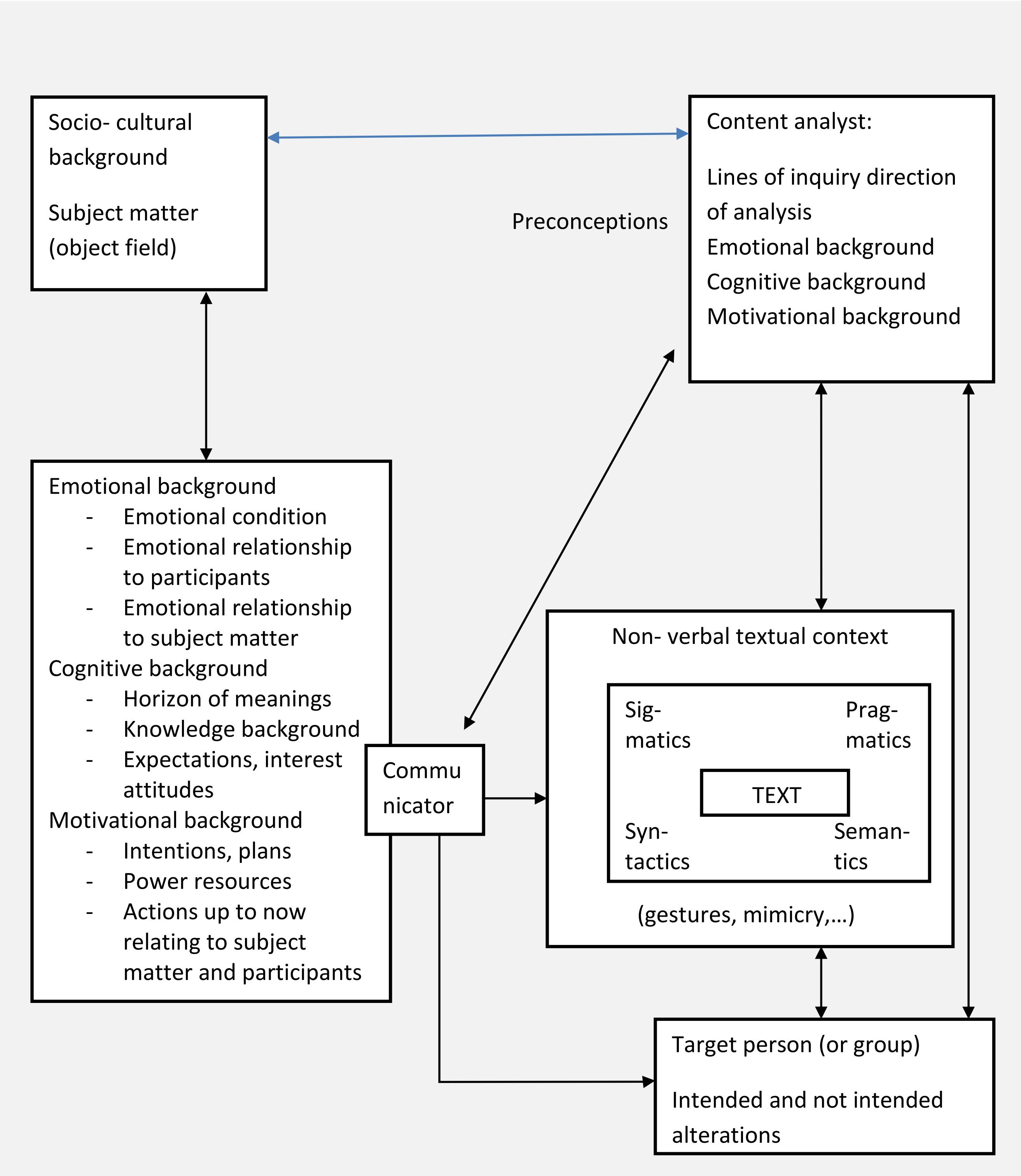


Figure 8: Content-analytical communication model

In this extended model we can now distinguish quite varied directions that a content analysis might take:

- One aim is to arrive at statements about the subject matter, above all in the case of document analyses.
- Content analyses in psychotherapy are mostly intended to bring out something about the emotional condition of the communicator.
- In literary studies the chief aim is usually to analyze the text for its own sake, with the socio-cultural background as the context.
- American propaganda research during the Second World War aimed at using content analyses to define the intention of the communicator.
- Analysis of the mass media frequently attempts to arrive at statements about their effects on the public, the target group, that is.

4.5 Content-Analytical Units

It is a central element of content-analytical procedures that the text is not interpreted as a whole but divided into segments. The categories are assigned to segments of text. This segmentation has to be defined in advance. Only if the segmentation rules, which are called units of analysis within the content analysis, are explicit, a second coder can come to similar results. This segmentation is important on three levels: First is has to be decided, how sensitive the analysis should be. Is it sufficient to detect slight undertones in the text to code it or are complete words, sentences or paragraphs necessary? The second decision is how many materials are relevant to come to a coding decision. And the third segmentation concerns the portions of text which are confronted with the category system.

Quantitative content analysis differentiates the following units (cf. Krippendorff, 1980), which are important for qualitative content analysis as well:

- The **coding unit** determines the smallest component of material which can be assessed and what the minimum portion of text is which can fall within one category.
- The **context unit** determines the largest text component, which can fall within one category.
- The **recording unit** determines which text portions are confronted with one system of categories.

The recording unit sometimes is called “unit of analysis”. But this is maybe confusing, because all three are units of analysis. Other sources call it “unit of enumeration”, but this will make more sense in contexts of quantitative content analysis.

The definition of these units is important for the intersubjectivity of the procedures, especially when inter-coder agreement tests are intended. If two coders refer to different content-analytical units, the agreement test is unfair.

Link to QCAmap software (www.qcamap.org):

In QCAmap you are forced to define the content-analytical units. If you leave this open a coding of the text is not possible.

Inductive category development (cf. chapter 6.2), one of the most common procedures of Qualitative Content Analysis, formulates categories and step-by-step augments the categories working through the text. At the end the category system stands for the whole material, so the recording unit has to comprise all text material for analysis.

> Link to QCAmap software (www.qcamap.org):
>
> In QCAmap, choosing inductive category development, the recording unit (all texts) is already fixed as default and cannot be changed.

In deductive category assignment the recording unit could be persons (in an interview study) or documents (issues in a newspaper analysis e.g.). The result of the content analysis will be one coding decision for each recording unit.

The coding unit expresses the sensitivity of the analysis. Is a slight overtone within one word (seme) sufficient for a coding decision, or should it be a complete phrase? You could use the linguistic terms mentioned in chapter 3.4 for defining the coding unit:

- Seme
- Phoneme
- Syllable
- Word
- Phrase
- Paraphrase
- Clause
- Sentence
- Proposition
- Paragraph
- Page

The context unit can be the same as the recording unit; but often it is broader. Even if the recording unit is only the answer to a specific interview question, the context unit could be established as the whole case. Sometimes there are additional observations during interviews or focus groups, transcribed in an observation protocol. Or there is further information about the persons or their cultural or social background which all could be made part of the context unit.

4.6 A General Step-by-step Model of Qualitative Content Analysis

In the next step the main consideration is to determine the special technique(s) of this analysis (see the following chapter) and to construct a procedural model for the analysis. The strength of Qualitative Content Analysis relative to other interpretation methods resides precisely in the fact that the analysis is resolved into individual steps of interpretation which are determined in advance. The whole process is thereby made comprehensible to others and intersubjectively testable; therefore it can also be transferred to other subjects, is available for use by others and can be regarded as a scientific method.

The procedural model for the analysis must certainly be adapted to suit the particular material and the specific problems concerned in particular cases. However, it is possible to construct a general model for orientation. The first stages of analysis in this model (figure 9) we have just discussed in chapter 5.2 to 5.4. For the next steps it is necessary first of all to establish **units of analysis**, in order to raise the level of precision of the content analysis.

The general procedural model is then the following (Fig. 9):

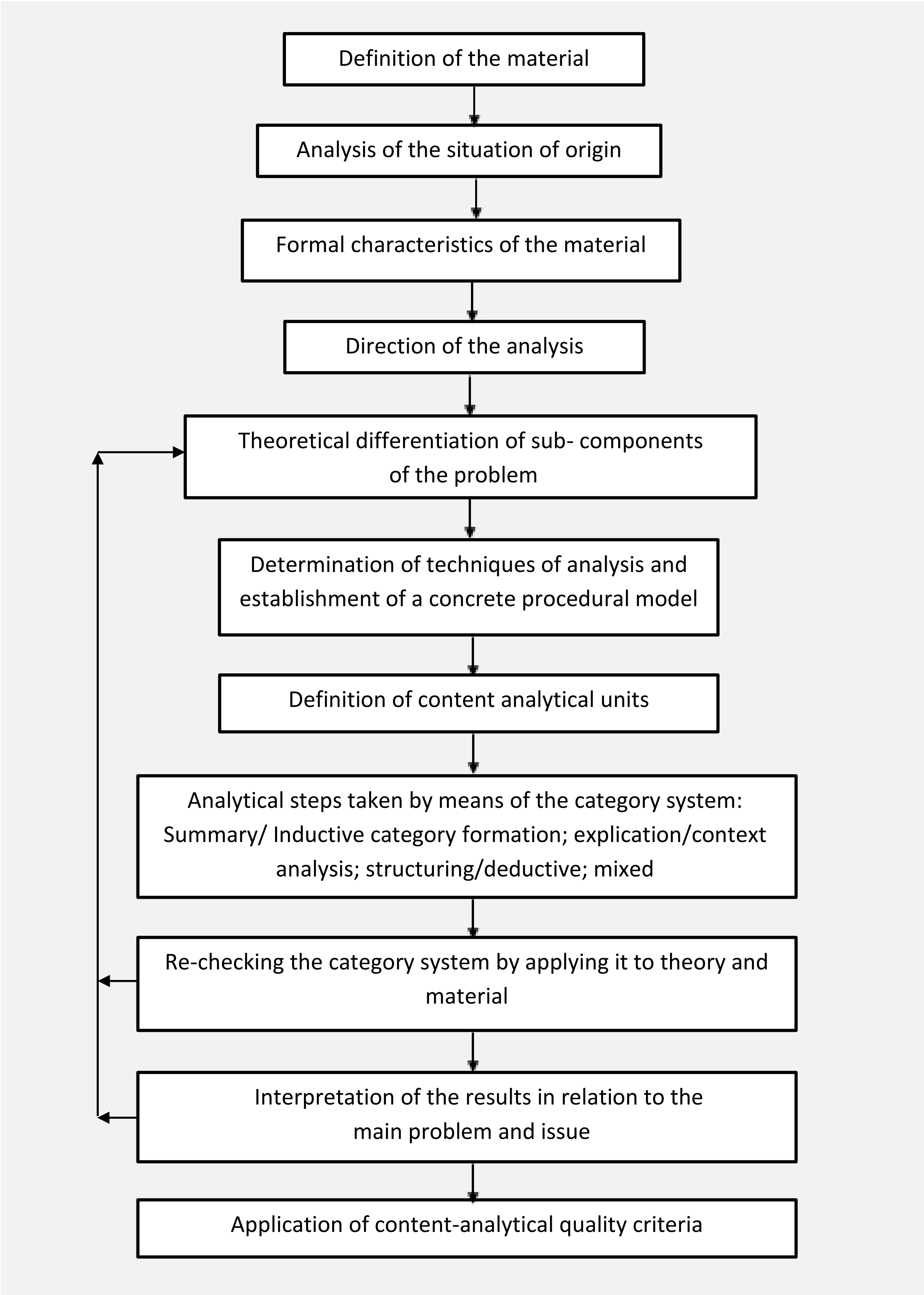


Figure 9: General content-analytical procedural model

5. Example

5.1 Presentation of the Corpus Material

Within the framework of a project fostered by the DFG (German Society for Scientific Research), and entitled "Cognitive control in crisis situations: unemployment among teachers", open-ended interviews were conducted with jobless teachers. How does the individual experience this situation, what stresses and strains does he or she feel in which particular areas, how does he view his particular position, how does he cope with it inwardly, and what attempts does he make to deal with it outwardly? These questions were put to a random sample of 75 unemployed teachers who were each interviewed seven times in the course of one year. Stress patterns and coping procedures were to be examined also with reference to the biography and life experience of the particular individual concerned. To this end, questions were also asked about the first removal from the parental home, initial teaching experiences during undergraduate practical training phases, experiences during postgraduate training, and experience of the final examination, the Second State Examination for Teachers.

The interviews were tape-recorded and then transcribed as typescripts. These scripts have a total length of nearly 20,000 pages, and were analyzed using content analytical procedures.

Four samples taken from the interview section on postgraduate training will be considered in the following. The interviews are found in the appendix.

5.2 Defining the Text Material

Content analysis is a method of data analysis, i.e. it concerns language material which already exists in a finished form. In order to decide what can be interpreted at all from the material, it is necessary for an exact analysis of this base material to be carried out right at the beginning. This procedure, known in the historical sciences as source study or source evaluation, is all too often overlooked or neglected in content analysis.

Basically three stages of analysis must be distinguished here:

5.2.1 Determining the Material

First of all the material on which the analysis is to be based must be defined exactly. This "corpus" should not be extended or altered during the analysis unless certain conditions occur which render it vitally necessary.

In many cases a selection from a larger volume of material must be made. Problems of sample selection thereby come to the fore (cf. on this point Krippendorff, 1980, Ch. 6). Here, attention should be paid to the following points:

- that the basic volume of corpus material is exactly defined in its entirety;
- that the body of selected samples is established according to considerations of economy and representativeness;
- that finally the samples are taken according to a certain model (purely random selection; selection according to quotas established in advance; stratified or cluster selection).

The script passages selected from the DFG project "Teacher Unemployment" concern four case study examples from the first batch to be examined, each of them, respectively, from the first round of interviews. With all of them the interview passage selected is the one, in which questions are being asked on first practical experiences of teaching during postgraduate training. The main motive for choosing these examples was the clarity and vividness of the material, which cannot be viewed as representative.

The individuals involved are:

Case A: high school teacher (male) of physics and geography

Case B: high school teacher (male) of physical education and geography

Case C: high school teacher (male) of physical education and geography

Case D: high school teacher (female) of English and history

All four passed the state examination but were not employed by the state education service owing to the lack of scheduled positions vacant at the time. The interview participants were obtained via the German teacher union (GEW) and were approached directly by the interviewer.

5.2.2 Analysis of the Circumstances of Origin

An exact description is required of where, from whom, and under what conditions the material originated. The following is particularly important:

- the author of the material and/or the parties involved in its production;
- the emotional, cognitive and motivational background of the author(s);
- the target group for which the material is intended;
- the concrete circumstances of origin;
- the socio-cultural background.

In respect to our example: Participation in the interviews was voluntary. A certain reciprocal effect was brought about by the fact that the interviewers on their part placed an advisory folder containing collated information on employment chances, application possibilities, alternative professional opportunities etc. at the disposal of the participants. The conversations are of two kinds: half-structured interviews (in which the interviewer has a guide matrix of questions, the phrasing and sequence of which, however, he may vary); open-ended interviews (i.e. the interviewee can respond to the questions quite freely). The interviews were carried out by the author as part of the research project. They were held at the homes of the interviewees.

5.2.3 Formal Characteristics of the Material

Finally it is necessary to describe the form in which the material exists. As a rule, content analysis requires a written text as a basis. Such a text, however, does not necessarily have to have been written by the author himself. The "core text" forming the basis of the analysis often has further information added to it. This is usual above all with spoken language, when for instance during interviews or group discussions observational data is frequently incorporated into the script. Spoken language, mostly in tape-recorded form, must be transcribed. For this operation there are various transcription models (cf. chapter 4.3) which, even at this stage, can alter the original material considerably. These transcription rules must be defined exactly.

In respect to our example: The interviews were recorded on tape and then transcribed in typed form. The following instructions were given to those carrying out the transcription:

Research Project "Teacher Unemployment"
Institute for Education and Educational Psychology, University of Munich

Instructions for interview transcription

60 machine strokes per line
38 lines, interval 1.5

* Please transcribe completely and verbatim (leaving incomplete portions and repetitions just as they are).

* The content should come first, however: "er" and similar phonetic fillers can be left out; regional accents should be ignored and all standard words written in standard German. Genuine dialect expressions, however, are to be retained and transcribed according to accoustic perception.

* Indistinct passages should be marked by a row of dots (....) corresponding to the length of what was not discernible, so that the interviewer can add the missing sections subsequently.

* In the case of pauses, hesitations, etc., use a dash (-) with longer pauses several dashes. If the reason for the pause is evident, please give this in brackets.

* State other noticeable concomitants (such as laughter, throat-clearing, etc.) also in brackets.

* All other non-verbal features important for interpreting the content should also be stated in brackets, e.g.:

Interviewee: Hmnm (in agreement).

* Typing errors should be simply crossed through (xxxx). Do not use correction fluid or similar devices.
(Irrelevant when transcribed on PC!)

* We require the original with two carbon copies. (Irrelevant wehen transcribed on PC!) The material can be obtained from us.

* The format is 60 machine strokes per line, interval 1.5, 38 lines per page, cf. boxed portion of these notes.

* When the interviewer askes a question, or simply speaks, please place the symbol "Q" (for "question") right at the edge of the margin, then a colon followed by <u>two</u> spaces. If more than one line is spoken, please begin the next lines right at the edge of the margin.

* When the interviewee, i.e. the unemployed teacher, is speaking, please use the symbol "T" (for "teacher")

* In the case of any further questions do not hesitate to contact us at any time.
We wish you and us a fruitful collaboration.

Figure 13: Notes on interview transcription for the research project "Teacher unemployment"

5.2.4 Direction of Analysis

The project from which the material is taken is oriented towards developmental psychology. The interviews were intended to encourage participants to report on their current feelings, their cognitive management of the situation, their coping efforts hitherto, and those further planned to deal with the situation, and on their own biographical experiences. According to the content-analytical communication model (cf. Figure 8), the direction of analysis is thus to use the text in order to arrive at statements on the emotional, cognitive and activity background of the interviewees.

5.2.5 Theory-oriented Differentiation of the Problem

Content analysis, according to our definition, is characterized by two features: rule-bound procedure (which will be dealt with in the next section) and the theoretical orientation of the interpretation. This is expressed first of all in the fact that the analysis follows a precise and theoretically based issue of substance. In this respect it is necessary to say something about the concept of theoretical orientation, as among those who favor the qualitative approach there is a negative attitude towards theory, which repeatedly asserts itself. It is frequently alleged that theories distort the material, constrain the view of the analyst and hinder “wholehearted immersion in the material”. However, if theory is understood as a system of general principles on the subject to be examined, then it constitutes nothing more than the cumulative experience of others in the same field. Theoretical orientation means, then, the tapping of this experience in order to achieve an advance in knowledge. What this entails concretely is that the issue in the focus of analysis must be defined precisely in advance, viewed within the context of current research on the topic, and as a rule divided into sub-issues. As far as our example is concerned, this means the following:

5.2.6 Theoretical Differentiation of Sub-issues

The sample material contains statements by four unemployed teachers on their experiences during the postgraduate phase of their teacher-training program. The literature on teacher training hitherto has indicated that this postgraduate training phase means for teachers previously educated in the almost exclusively theoretical atmosphere of a university a kind of shock effect ("professional practice shock" or "job strain") on being confronted with the realities of school life. (cf. Smagorinsky et al., 2004; Mueller-Fohrbrodt, Cloetta & Dann, 1978; Dann, Mueller-Forbroth & Cloetta, 1981).

This is accompanied by a change of attitude in the direction of a controlling, disciplinary and authoritarian stance towards school students, a concept of giftedness which stresses the hereditary limits to the fostering of students' talents, increased punitive and pressurizing behavior towards students, and a decreased level of professional involvement.

It is of interest in this connection to establish whether the experiences of *unemployed* teachers are similar. What was particularly examined in the DFG project was how far their interest in the teaching profession is influenced and how this affects the way they deal with their own unemployment situation.

A further point of analysis was the question of whether these experiences had influenced generalized control expectation (cf. Rotter, 1966) and the self-confidence of the individual, and had had effects on his current coping strategies.

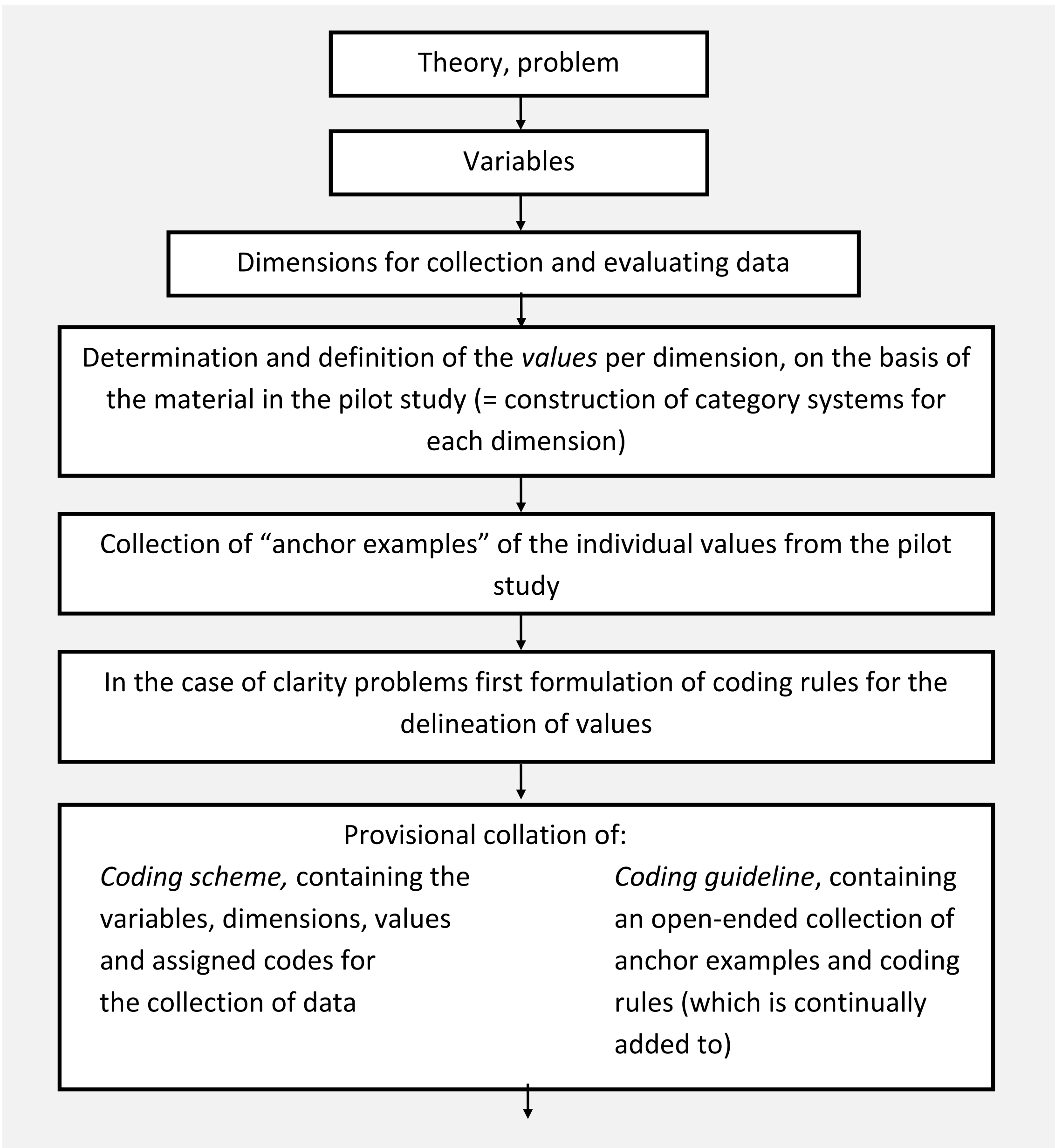
Two main questions emerge from this in relation to the sample material:

Question One: What are the main experiences of unemployed teachers with "professional practice shock"?

Question Two: What can be concluded from these experiences about the effects on self-confidence?

The next step in the general content-analytical step model (cf. Fig. 8) would be the determination of the specific content-analytical procedure. We have developed for Qualitative Content Analysis a set of different procedures, which now will be described. The example will be seized again for each technique.

Now back to our example: In the initial sections of this chapter we described the procedural model for the example analysis, which is to be used to demonstrate the various techniques in the next chapter; it will be continued during description of the individual techniques. In this way it is intended here to demonstrate the evaluation model of the whole project from which the sample material is taken (cf. Ulich et al. 1985). The core of this is a structuring content analysis or deductive category assignment (cf. Ch. 6.5), in which quantitative steps, extending to statistical analysis by electronic data processing, are incorporated. In addition, however, other purely qualitative content-analytical procedures are also employed for the analysis of non-systematically evaluated aspects.



Determining of content-analytical analysis units:

- Coding unit (min.): proposition
- Context unit (max.): all material from the respective case
- Recording unit: the respective case

Trial encoding by all five members of the project group from the first three interviews of the main inquiry phase:

1) Designation of the discovery points with colour markers according to the variables (a direct run-through of the material with all category schemes is impossible owing to the volume of material)
2) Coding (filling the coding scheme)

Revision of the coding scheme and coding guideline:

1) Values per dimension (categories) are discarded where they are too detailed and added where necessary.
2) Definitions of categories are made more precise.
3) Where there are discrepancies and problems of categorization in the sense of inexact delimitation of the values, an appropriate categorization is discussed and decided upon. On the basis of these cases new coding rules are formulated and incorporated in the coding guideline.
4) As far as they are important for the definition of the values, the coded portions of the interviews are incorporated in the coding guide as anchor examples.

Final version of the coding guideline; copying

Re-coding of the first three interviews according to the revised coding scheme

Testing of the *inter-coder reliability* and extension of the coding guideline: Further interviews from the main phase are coded by all five members of the project group according to the revised coding scheme; determination of inter-coder reliability; coding rules made more precise and new anchor examples incorporated in coding guide if there are discrepancies.

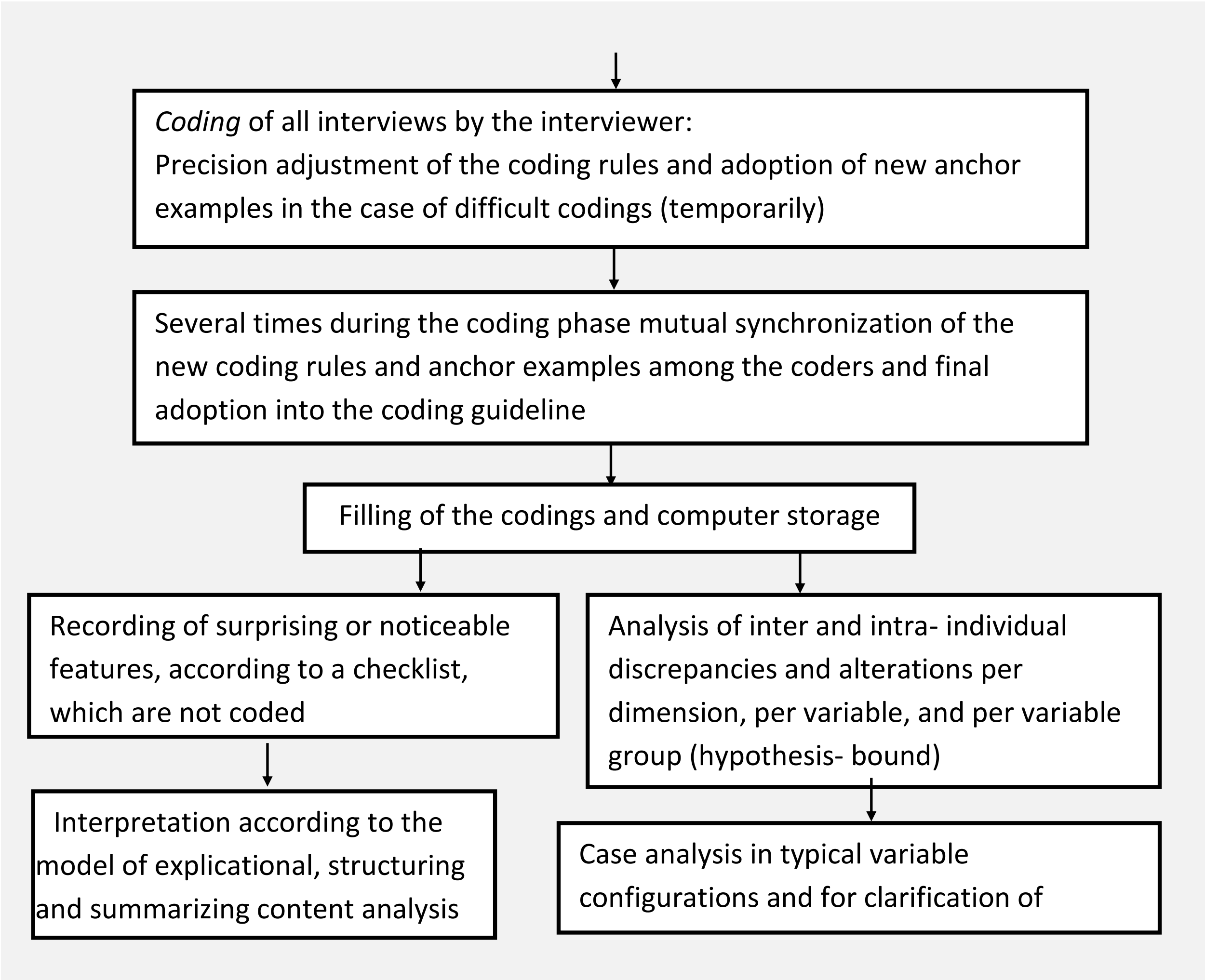


Figure 11: Step model for research project "Teacher Unemployment"

The example will be continued in the next chapters demonstrating the different procedures.

6. Specific Techniques of Qualitative Content Analysis

As already emphasized, qualitative content analysis is not to be conceived of here as an alternative to quantitative content analysis. The concern of this work is to develop methods of systematic interpretation which are applicable to the qualitative components necessarily involved in every content analysis, systematizing and making them testable through stages and rules of analysis. Quantitative procedures can certainly be incorporated into such an "interpretational theory", but then they simply occupy a new position. The concept "qualitative content analysis" may only be partly applicable to this approach, but will nevertheless be retained, in order to make the main bias clear and explicit.

In this chapter we propose concrete techniques of qualitative content analysis and demonstrate them with an example in the next chapter.

The aim of this book is to describe techniques of qualitative content analysis as basic procedural methods of systematic, i.e. theory- and rule-bound, textual understanding and textual interpretation.

The point of approach here is to find out the basic structure of ways in which texts are dealt with, both on an everyday informal level and on a scientific one. It is precisely this that is neglected by quantitative methods, which apply cut-and-dried procedures to the material without testing the assumptions implicit in them. This too must therefore be part of the approach of qualitative analysis.

6.1 Basic Forms of Interpretation

I would like to begin with the techniques and approaches which have been described above. It will be our task to emphasize what the analysis does with the material and what the role of interpretation is. These characterizations of interpretation type will then be categorized in fundamental interpretation procedures.

It could be shown that existing techniques of interpreting text material systematically are in their basic structures not so very different from one another and can be traced back to a few fundamental methods. The point of departure is mostly the individual text component which must be analyzed more exactly (for instance as regards to its textual context), evaluated in a certain direction, examined in its relations to other textual components (as a rule for the purpose of revealing textual structures) and often some kind of summary of the material is aimed at. So it seems to me that we can differentiate between three fundamental forms of interpreting: summary, explication, and structuring. They can generally be described as follows:

Summary: The object of the analysis is to reduce the material in such a way that the essential contents remain, in order to create through abstraction a comprehensive overview of the base material which is nevertheless still an image of it.

Explication: The object of the analysis is to provide additional material on individual doubtful text components (terms, sentences...) with a view to increasing understanding, explaining, interpreting the particular passage of text.

Structuring: The object of the analysis is to filter out particular aspects of the material, to give a cross-section through the material according to pre-determined ordering criteria, or to assess the material according to certain criteria.

These three basic forms of interpretation correspond also to the everyday view of the basic methods which can be employed in order to analyze (language) material as yet unfamiliar. At this point I would like to perform a little experiment in mind:

> Imagine that in the course of a hike across open country I suddenly come face to face with a gigantic piece of rock (perhaps a meteorite or the like). Supposing I wanted to find out what this thing was that was confronting me. How could I proceed?
>
> First I would retreat to a nearby place of high ground from where I could view the rock in its entirety. From this distance, certainly, I would no longer be able to see details, but I would have the whole object in its general rough outline before me, effectively in a reduced form (summary).
>
> Then I would go right up to the rock again and look at portions of it more closely which seem particularly interesting. I would break pieces off and examine them (explication).
>
> Finally I would try to break the whole rock open in order to get some idea of its internal structure. I would try to identify individual components, to take measurements of the rock, ascertaining its size, hardness, and weight by carrying out various measuring operations (structuring).

The most varied mixtures of these analysis types are of course possible, but the development of qualitative techniques should first of all take the basic forms as its point of departure.

These basic forms, however, must be further differentiated before an exact description of procedure is possible. Beside usual summaries the same procedures are useful for inductive category formation; a criterion for the categories is defined and aspects to this criterion are stepwise gathered in the material. Forms of explication are possible which use the textual context for the elucidation of a particular text passage (narrow contextual analysis); however, the most common method of hermeneutical interpretation is to use further material beyond the textual context for explication (broad contextual analysis). With structuring too, sub-groups must be distinguished: the structuring categories can form an ordinal scale or can remain as nominal categories. And mixed procedures with inductive and deductive steps of analysis (e.g. theme analysis, typological analysis) should be conceptualized as well.

Through this differentiation we arrive at nine distinct forms of analysis:

Reduction	(1)	summarizing
	(2)	inductive category formation
Explication	(3)	narrow contextual analysis
	(4)	broad contextual analysis
Structuring	(5)	nominal deductive category assignment
	(6)	ordinal deductive category assignment
Mixed	(7)	content structuring/theme analysis
	(8)	type analysis
	(9)	Parallel forms

This catalogue of qualitative analysis techniques is to be understood as a first approach and does not claim to be complete. However, it can serve as a starting point for systematic testing and further development. Qualitative content analysis aims, then, to develop these nine forms of analysis through differentiation into individual analytical steps and the formulation of interpretation rules concerning systematic content-analytical techniques.

6.2 Summarizing

The first two techniques try to reduce the material to core contents or aspects.

It is in the development of individual analytical steps for summary that one can rely largely on the support of previous studies. The psychology of text comprehension (Van Dijk, 1980; Ballstaedt, Mandl, Schnotz & Tergan, 1981) has described exactly how summaries usually proceed in everyday life. Central points are the distinction between ascending (text-bound) and descending (pattern-bound) processing and the formulation of macro-operators for reduction (see chapter 3.4).

The basic principle of a summarizing content analysis is then that the level of abstraction of the summary should be exactly determined in each case, so that the macro-operators can be used to transform the material precisely to that level. This level of abstraction can now be generalized upon gradually; the summary becomes increasingly abstract. A general content analytical process model for summarizing can therefore be diagrammed as follows:

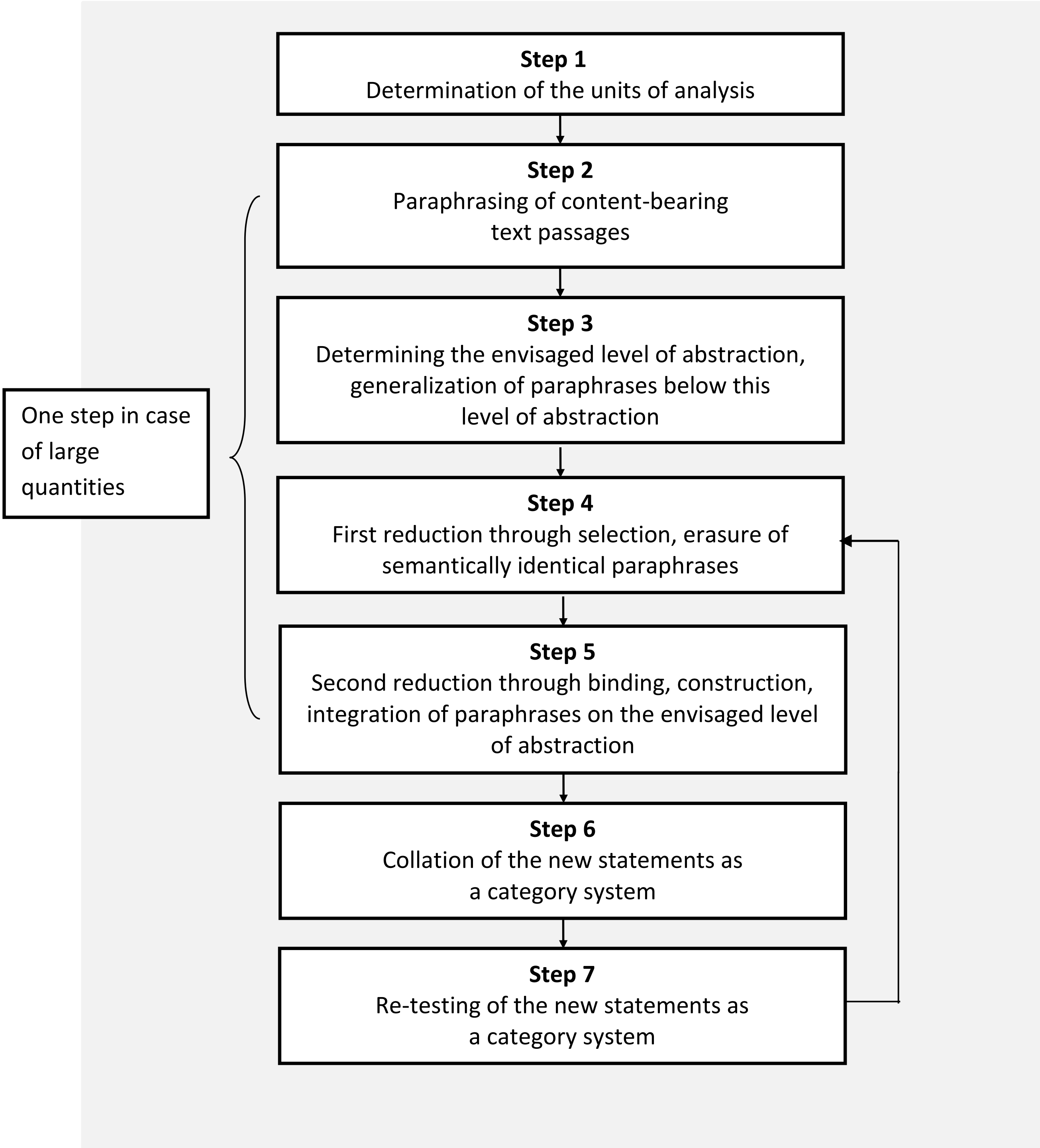


Figure 12: Step-by-step model of summarizing content analysis

The first steps, then, address themselves to describing the material exactly and determining what is to be summarized in the light of the problem involved. After this the analysis units must be determined (cf. chapter 4.5).

The individual coding units are now re-written in a short descriptive form which is confined to the content (paraphrasing). At this stage already, embellishing text components which add nothing to the content are omitted. The paraphrases should be formulated on a uniform stylistic level. This is important especially when several different speakers are involved (e.g. in a group discussion). The final version should be a grammatically reduced one (for instance, "Yes, you see, at the time I didn't really feel any strain, basically" becomes "no strain felt") (cf. the S1 rules on the next page). Where the volume of material is not that large, these paraphrases are actually written in full; where this would be too complex or work intensive, the next two steps of analysis are applied simultaneously.

In the next step the intended level of abstraction of the first reduction is determined according to the nature of the material. All paraphrases below this level must now be subjected to generalization (generalizing macro-operator). At this point, as well as during further stages of reduction, cases of doubt must be resolved with the help of theoretical preconceptions. Paraphrases above the intended abstraction level are initially left as they are (cf. the S2 rules). This produces a few content-identical paraphrases which can now be cut. Similarly, insignificant and vague paraphrases can be omitted (omission and selection macro-operators) (cf. the S3 rules). In a second stage of reduction several paraphrases referring to one another and occurring passim throughout the material are summarized and expressed in a single new statement (binding, construction and integration macro-operators) (cf. the S4 rules).

At the end of this reduction phase exact checking must take place to ascertain whether the new statements collated as a category system really do still represent the base material. All original paraphrases from the first stages of treatment must be included in the category system. Even more thorough, of course, is a re-check of the summary by referring to the base material itself. The first run-through of the summary is now complete.

Often, however, a further summary is necessary. This is quite simple to carry out by raising the abstraction level higher still and re-applying subsequent interpretation steps. The result of this process is a new, more general and more brief category system, which again must be re-checked. This cyclical process can be applied repeatedly until the result corresponds to the intended reduction of the material.

If the volume of material is large, it is often impossible to paraphrase all the content-relevant parts of the text. In this case several analysis steps can be brought together as one. The text passages are then paraphrased to the intended abstraction level from the beginning. Before each new generalized paraphrase is written out, checks are made to ensure whether it is not included in those that have been made already, or related to them, so that it could be bound or integrated with them to form a new statement.

From this description of the model and the account of the above described macro-operators we can now draw up interpretation rules for the summary form of qualitative content analysis. They are related to the four points in the process at which the material is reduced:

S1: Paraphrasing

S1.1 Cut all the text components which are not content-bearing or only minimally so, such as embellishing, repetitive, or explanatory expressions.

S1.2 Transpose the content-bearing parts of the text on to a uniform stylistic level.

S1.3 Transform them into a grammatically abbreviated form.

S2: Generalization to the required level of abstraction

S2.1 Generalize the referents of the paraphrases to the defined level of abstraction, so that the old referents are implied in the newly formulated ones.

S2.2 Generalize the sentence kernels (predicates) in the same way.

S2.3 Leave those paraphrases standing which are above the intended level of abstraction.

S2.4 In cases of doubt make use of theoretical preconceptions.

S3: First reduction

S3.1 Cut semantically identical paraphrases within units of evaluation.

S3.2 Cut paraphrases which are not felt to add substantially to the content on the new level of abstraction.

S3.3 Adopt the paraphrases which continue to be thought of as vitally content-bearing (selection).

S3.4 Resolve cases of doubt with the aid of theoretical preconceptions.

S4: Second reduction

S4.1 Combine paraphrases with identical or similar referents and similar statements to form one paraphrase (binding).

S4.2 Combine paraphrases with several statements on the same referent into one (construction/integration).

S4.3 Combine paraphrases with identical or similar referents and differing statements into one paraphrase (construction/integration).

S4.4 Resolve cases of doubt with the aid of theoretical preconceptions.

Link to QCAmap software (www.qcamap.org):

Summarizing will be implemented within the software package in autumn 2014. The program leads you through the steps of analysis. A special screen is offered for the tabulation of paraphrases and reductions.

Example

For a demonstration of the summary form of qualitative content analysis using our sample material, the first central question is very suitable (cf. p. 59): "What are the main experiences of the unemployed teachers with ′practice shock′?" The remarks of the four teachers on "practice shock" which take up 11 pages of the appendix (p. 125-135) will now be summarized in two reduction operations to a length of half a page.

The first thing to be made clear when determining the units of analysis is that with the summary form the recording unit and the context unit always coincide. In the case of our example this unit is in the first operation the individual case, and in the second the entire material. The coding unit, however, is conceived of more narrowly. This determines the units which form the basis of the summary as paraphrases in the first run-through of the material. In the example the coding unit is every complete statement by a teacher on experiences, assessment and effects of the postgraduate training phase compared with the theoretical part of the course at university.

In the following the first reduction operation will be described. The case number and page reference of the respective text passage is the first information to be given in the table. In the next columns the paraphrases of the content-bearing text passages are then portrayed and numbered consecutively.

The abstraction level of the first reduction run-through was determined as follows: statements relating to the postgraduate training phase in a form as general as possible, but case-specific ones (per teacher); in other words, statements by the teacher concerned about his entire postgraduate phase which summarize his experience of "practice shock".

In the center main column the individual paraphrases have been generalized to this abstraction level. Double statements, or insignificant ones, were eradicated for this column.

In the final column the remaining statements have been combined into new ones for each case through binding, integration and construction, and constitute the result of the first run-through. As they were the first category system, they were numbered.

Case	page	Paraphrase	Generalization	Reduction
A	125	P1: No psychological strain experienced through practice shock	No practice shock experienced as very enjoyable because	K1: Practical teaching not experienced as a shock, but as very enjoyable, because - previous teaching experience - country school without discipline problems - had no unrealistic expectations - had good relations to students
A	125	P2: On the contrary, was very keen on teaching practice	Tended to look forward to teaching practice	
A	125	P3: University = purely academic course, little to do with teaching	At university teaching experience not part of course	
A	125	P4: Was able, however, to gather teaching experience beforehand	Prior experience of teaching	
A	125	P5: Practice was very enjoyable	Practice enjoyable	
A	125	P6: As far as subject matter was concerned, teaching was simple and fascinating for the students	Easily teachable subject matter as a condition	K2: Without these conditions practice shock undoubtedly conceivable
A	125	P7: Had been waiting to begin teaching with some impatience	Had looked forward to starting to teach	
A	125	P8: But there are some dis-appointments about pupils not being what one thinks they should be	Disappointments too	
A	126	P9: Certainly not a practice shock	No practice shock	
A	126	P10: Workload not so heavy (at most in a branch of a school)	Low workload	
A	126	P11: Frustration of teacher at inner city school with possible discipline problems among students possible	Frustration of teacher at inner city school	
A	126	P12: Own efforts compensated for by enjoyment of teaching	Found the work enjoyable	
A	126	P13: Students still like me there	Had good relations to students	
A	127	P14: Am too realistic to have had wrong ideas about teaching	No unrealistic expectations	
A	127	P15: With 35 students and the amount of subject matter involved opportunity for educational work in any case low	Possibilities for educational work only low	K3: No practice shock, owing to flexibility, realistic attitude, adaptability and conversations with open colleagues

B	128	P16: No personal direct experience of practice shock	No practice shock	K4: Belief in getting by without disciplinary measures, just on the
B	128	P17: Positive "Here I come!" type of attitude at the outset	~~The feeling of being able to do it better at the beginning~~	
B	128	P18: Was even criticized for my teaching by another student teacher	The feeling of being able to do it better even with other students	
B	128	P19: Told him the "persuasive" method possible only in the rarest of cases	Illusion, as the "persuasive" method possible only in the rarest cases	strength of persuasion, an illusion, because - even experienced teachers have difficulties
B	128	P20: At the beginning I also said, "That can be done differently."	~~The feeling of being able to do it better at the beginning~~	- students expect disciplinary measures - large classes
B	128	P21: After some initial difficulties, managed to achieve a good relationship with my first class	Good relationship achieved with the class	- frequent change of class - relativity of educational values - good relation to
B	128	P22: Was not shocked	~~No practice shock~~	students is also possible on a different basis
B	128	P23: Took it as it came	Realistic and adaptable	
B	128	P24: Experienced teachers have the same problems, so no need to feel al failure	No feeling of personal failure, as other teachers also have problems	K5: Ski trips/sport/games can compensate for harsh image
B	128	P25: Few teachers admit their difficulties	Few teachers admit their difficulties	K6: Dilemma of trying out pedagogical behavior types
B	128	P26: Fellow teachers open and communicative		and nevertheless remaining consistent
B	128	P27: Talking to colleagues as the best solution to practical problems	Talking to colleagues as the best solution to practical problems	
B	128	P28: Not directly shocked	~~No practice shock~~	
B	128	P29: Am very flexible and always know how to react	Am flexible	
B	128	P30: Easy to talk about educational values with the benefit of hindsight	Educational values always controversial	
B	128	P31: Shouting often more useful than trying hard to persuade	~~Shouting often more useful than trying hard to persuade~~	

B	128	P32: With large classes often forced into doing questionable things	Large classes make pedagogical behavior difficult	
B	128	P33: Students want something done	Students want measures taken	
B	129	P34: Could never imagine doing such a thing	An illusion to imagine getting by without disciplinary measures	
B	129	P35: One acquires a catalogue of possible reactions to discipline problems	One acquires discipline catalogue	
B	129	P36: One should try out different methods during postgraduate training	One should try things out	
B	129	P37: Have tried "banging on the table" and it has had short-term effects	Have tried disciplinary methods successfully	
B	129	P38: Tried out tips like this, worked on myself	Have tried disciplinary methods successfully	
B	129	P39: This must be pushed through, because the class allows no retreat	Pressure to be consistent	K7: Practice shock a great problem owing to obligation to adapt to ideas of seminary instructors in order to acquire good grades; gnaws at self-confidence and own ego
B	129	P40: That is a dilemma	Caught between experimentation and consistency	
B	129	P41: A lot learnt about behavior towards students	Learnt how to deal with students	K8: Perhaps due to - greater sensitivity - not a grade-one candidate - not a "conferencier" type - not very adaptable
B	129	P42: Had good relations with students	Had good relations with students	
B	129	P43: On school skiing trips, and often in games classes too, one has a completely different relationship with students	Ski trips/games classes different relationship	
B	129	P44: Geography more difficult, as fewer hours of lessons	Difficult when fewer lesson hours	
C	130	P45: Practice shock as a great problem	Practice shock as a great problem	
C	130	P46: Dependency on seminary teacher initially dominant	Dependency on seminary teacher	

C	130	P47: First of all viewed classes as gloomy affairs, as it all could be done differently	Initially the feeling that it could be done differently
C	130	P48: These ideas cannot be realized during postgraduate training	This is not realizable
C	130	P49: One wants to be assessed as positively as possible	Dependency on evaluation of performance
C	130	P50: That causes conflict	Causes conflict
C	130	P51: Anything the seminary teacher feels to be inappropriate cannot be done	Pressure to conform to seminary teacher
C	130	P52: One has to conform to the seminary teacher from the outset	~~Pressure to conform to seminary teacher~~
C	131	P53: Am not the type to run through schematic rules immediately	Not the type to solve all problems schematically
C	131	P54: When one seeks relation-ships to students reactions often occur in one which do not conform to official stipulations	Own ideas often deviant
C	131	P55: In this one is frequently wrong in one's assumptions	Often false ideas
C	131	P56: It may be that I am more than usually sensitive in that direction	Much more sensitive
C	131	P57: Other teacher trainees have seen it that way too, though	Others feel the same way
C	131	P58: Permanent awareness of the need to get as good a grade as possible	~~Pressure for good assessment from seminary instructor~~
C	131	P59: People try for all they're worth to get as good a grade as possible	~~Pressure for good grades~~
C	131	P60: Pressure to conform	~~Pressure to conform~~